Emission of high-energy gamma rays, neutrinos and secondary electrons/positrons

(Lecture 7)

Satyendra Thoudam
Khalifa University

Outline of this lecture

- Emission from hadronic processes
  - Proton-proton interaction
    - Gamma-rays spectrum
    - Electrons spectrum
    - Neutrinos spectrum
  - Proton-photon interaction
    - Emission from photopion process
    - Emission from electron/positron pair production
- Emission from leptonic processes
  - Electron-positron annihilation
  - Bremsstrahlung emission
  - Inverse Compton interaction
  - Synchrotron radiation

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(1) Proton-proton inelastic collision:

$$p + p \rightarrow \pi^+, \pi^-, \pi^0$$

$$\pi^0 \rightarrow \gamma + \gamma \qquad \text{Mean lifetime } 1.78\times10^{-16} \text{ s}$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu \quad ; \quad \pi^- \rightarrow \mu^- + \bar{\nu}_\mu \qquad \text{Mean lifetime } 2.55\times10^{-8} \text{ s}$$

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu \quad ; \quad \mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu \qquad 2.20\times10^{-6} \text{ s}$$

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(1) Proton-proton inelastic collision:

$$p + p \rightarrow \pi^+, \pi^-, \pi^0$$

$$\pi^0 \rightarrow \gamma + \gamma \qquad \text{Mean lifetime } 1.78\times10^{-16} \text{ s}$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu \quad ; \quad \pi^- \rightarrow \mu^- + \bar{\nu}_\mu \qquad \text{Mean lifetime } 2.55\times10^{-8} \text{ s}$$

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu \quad ; \quad \mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu \qquad 2.20\times10^{-6} \text{ s}$$

*At high energies, the three types of pions are produced with similar probabilities.

*Neutrinos are smoking gun of hadronic processes.

*Neutrinos produced in flavor ratio: $\nu_e : \nu_\mu : \nu_\tau = 1 : 2 : 0$

*Propagation changed the flavor ratio: $\nu_e : \nu_\mu : \nu_\tau = 1 : 1 : 1$

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(1) Proton-proton inelastic collision:

$$p + p \rightarrow \pi^+, \pi^-, \pi^0$$

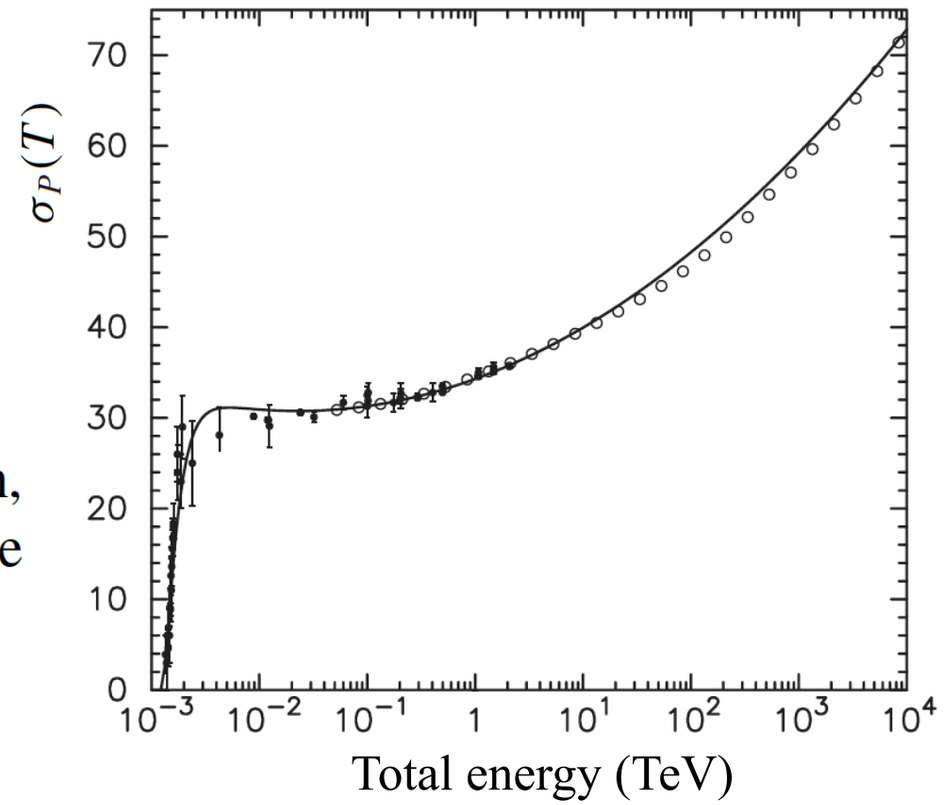
Total interaction crossection:

$$\sigma_P(T) = \left(34.3 + 1.88L + 0.25L^2\right)\left[1 - \left(\frac{T_{\rm th}}{T}\right)^4\right]^2 \text{ mb},$$

[1mb = millibarn =10^{-27} cm^2]

where T is the total energy of the cosmic ray proton, $L = \ln(T/1\text{ TeV})$ and $T_{\rm th} = 1.22$ GeV is the threshold energy for the production of π^0 mesons.

corresponds to kinetic energy of 282 MeV for the proton



Kelner, Aharonian, Bugayov, 2006, Phys. Rev. D, 74, 034018

Emission of high-energy gamma rays, neutrinos and secondary e±

Gamma-ray emissivity:

$$p + p \rightarrow \pi^0 \rightarrow \gamma + \gamma$$

*Emissivity $q_\gamma(E_\gamma)$: $\dfrac{\text{Number of particles}}{\text{volume x Energy x Time}}$ => $\dfrac{\#}{\text{cm}^3 \text{ x GeV x sec}}$

* Basically, it is the particle production rate per unit volume.

Emission of high-energy gamma rays, neutrinos and secondary e±

Gamma-ray emissivity:

$$p + p \rightarrow \pi^0 \rightarrow \gamma + \gamma$$

*Emissivity $q_\gamma(E_\gamma)$: $\dfrac{\text{Number of particles}}{\text{volume x Energy x Time}}$ => $\dfrac{\#}{\text{cm}^3 \text{ x GeV x sec}}$

* Basically, it is the particle production rate per unit volume.

* The intensity of gamma rays in a given direction can be calculated from emissivity:

$$I_\gamma(E_\gamma) = \frac{1}{4\pi} \int q_\gamma(r, E_\gamma) dr$$

where the integration is over the line-of-sight distance.

Emission of high-energy gamma rays, neutrinos and secondary e±

Gamma-ray emissivity:

$$p + p \rightarrow \pi^0 \rightarrow \gamma + \gamma$$

- The gamma-ray production involves a **two-step** process.
- Emissivity calculation will involve **two** integrals:
  - (1) Over the cosmic-ray kinetic energy.
  - (2) Over the π^0 total energy.

For details on the kinematics involved in the calculation, I recommend :
Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 110-122).

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Gamma-ray emissivity: * The π^0-mesons decay γ-rays emissivity is given by (and references therein)

$$q_\gamma^{\pi^0}(E\gamma) = 2\int_{E_{\pi^0}^{\min}(E_\gamma)}^{\infty} \frac{q_{\pi^0}(E\pi^0)}{\sqrt{E_{\pi^0}^2 - m_{\pi^0}^2 c^4}} \mathrm{d}E_{\pi^0}$$

where $E_{\pi^0}^{\min}(E_\gamma) = E_\gamma + m_{\pi^0}^2 c^4 / 4E_\gamma$ and m_{π^0} is the π^0-meson rest mass energy.

Thoudam, 2006, Astroparticle Physics 2006, 25, 328

Aharonian & Atoyan, Astronomy and Astrophyssics, 2000, 362, 937

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Gamma-ray emissivity: * The π^0-mesons decay γ-rays emissivity is given by (and references therein)

$$q_\gamma^{\pi^0}(E\gamma) = 2\int_{E_{\pi^0}^{\min}(E_\gamma)}^{\infty} \frac{q_{\pi^0}(E\pi^0)}{\sqrt{E_{\pi^0}^2 - m_{\pi^0}^2 c^4}} \mathrm{d}E_{\pi^0}$$

where $E_{\pi^0}^{\min}(E_\gamma) = E_\gamma + m_{\pi^0}^2 c^4 / 4E_\gamma$ and m_{π^0} is the π^0-meson rest mass energy.

* The emissivity of π^0 mesons is calculated as $q_{\pi^0}(E\pi^0) = 4\pi\, n_{\mathrm{H}} \int_{E_{\mathrm{p}}^{\min}(E_{\pi^0})}^{\infty} \mathrm{d}E_{\mathrm{p}} J_{\mathrm{p}}(E\mathrm{p}) \frac{\mathrm{d}\sigma_{\pi^0}(E_{\pi^0}, E_{\mathrm{p}})}{\mathrm{d}E_{\pi^0}}$

where $J_{\mathrm{p}}(E\mathrm{p})$ is the CR proton flux,

$\mathrm{d}\sigma_{\pi^0}(E_{\pi^0}, E_{\mathrm{p}})/\mathrm{d}E_{\pi^0}$ is the differential cross-section for a CR proton of kinetic energy E_{p} to produce a π^0 with total energy E_{π^0},

n_{H} is the target hydrogen density,

$E_{\mathrm{p}}^{\min}(E_{\pi^0})$ is the minimum proton kinetic energy that contributes to the production of a pion with energy E_{π^0}.

Thoudam, 2006, Astroparticle Physics 2006, 25, 328

Aharonian & Atoyan, Astronomy and Astrophyssics, 2000, 362, 937

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Gamma-ray emissivity:

* Combining the two equation in the previous slides, we have for the gamma-ray emissivity:

$$q_\gamma^{\pi^0}(E\gamma) = 8\pi\, n_{\mathrm{H}} \int_{E_{\pi^0}^{\min}(E_\gamma)}^{\infty} \frac{1}{\sqrt{E_{\pi^0}^2 - m_{\pi^0}^2 c^4}}\, \mathrm{d}E_{\pi^0} \int_{E_{\mathrm{p}}^{\min}(E_{\pi^0})}^{\infty} \mathrm{d}E_{\mathrm{p}} J_{\mathrm{p}}(E\mathrm{p}) \frac{\mathrm{d}\sigma_{\pi^0}(E_{\pi^0}, E_{\mathrm{p}})}{\mathrm{d}E_{\pi^0}}$$

γ-ray production term **Pion production term**

Thoudam, 2006, Astroparticle Physics 2006, 25, 328

Aharonian & Atoyan, Astronomy and Astrophyssics, 2000, 362, 937

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Gamma-ray emissivity:

* Combining the two equation in the previous slides, we have for the gamma-ray emissivity:

$$q_\gamma^{\pi^0}(E\gamma) = 8\pi\, n_{\mathrm{H}} \int_{E_{\pi^0}^{\min}(E_\gamma)}^{\infty} \frac{1}{\sqrt{E_{\pi^0}^2 - m_{\pi^0}^2 c^4}}\, \mathrm{d}E_{\pi^0} \int_{E_{\mathrm{p}}^{\min}(E_{\pi^0})}^{\infty} \mathrm{d}E_{\mathrm{p}} J_{\mathrm{p}}(E\mathrm{p}) \frac{\mathrm{d}\sigma_{\pi^0}(E_{\pi^0}, E_{\mathrm{p}})}{\mathrm{d}E_{\pi^0}}$$

γ-ray production term **Pion production term**

The differential cross-section can be approximated as $\dfrac{d\sigma_{\pi^0}(E_{\pi^0}, E_p)}{dE_{\pi^0}} = \sigma_p(E_p)\delta(E_{\pi^0} - k_\pi E_p)$

where $k_\pi \sim 0.17$ is the inelasticity (the mean fraction of kinetic energy of proton transferred to π^0).

Thoudam, 2006, Astroparticle Physics 2006, 25, 328
Aharonian & Atoyan, Astronomy and Astrophyssics, 2000, 362, 937

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Gamma-ray emissivity:

* Combining the two equation in the previous slides, we have for the gamma-ray emissivity:

$$q_\gamma^{\pi^0}(E\gamma) = 8\pi\, n_{\rm H} \int_{E_{\pi^0}^{\rm min}(E_\gamma)}^{\infty} \frac{1}{\sqrt{E_{\pi^0}^2 - m_{\pi^0}^2 c^4}} {\rm d}E_{\pi^0} \int_{E_{\rm p}^{\rm min}(E_{\pi^0})}^{\infty} {\rm d}E_{\rm p} J_{\rm p}(E{\rm p}) \frac{{\rm d}\sigma_{\pi^0}(E_{\pi^0}, E_{\rm p})}{{\rm d}E_{\pi^0}}$$

γ-ray production term **Pion production term**

The differential cross-section can be approximated as $\dfrac{d\sigma_{\pi^0}(E_{\pi^0}, E_p)}{dE_{\pi^0}} = \sigma_p(E_p)\delta(E_{\pi^0} - k_\pi E_p)$

where $k_\pi \sim 0.17$ is the inelasticity (the mean fraction of kinetic energy of proton transferred to π^0).

*For a power-law proton spectrum: $J_p(E_p) \propto E_p^{-\Gamma}$

*Gamma-ray spectrum at high energies closely follows: $q_\gamma^{\pi^0}(E_\gamma) \propto E_\gamma^{-\Gamma}$

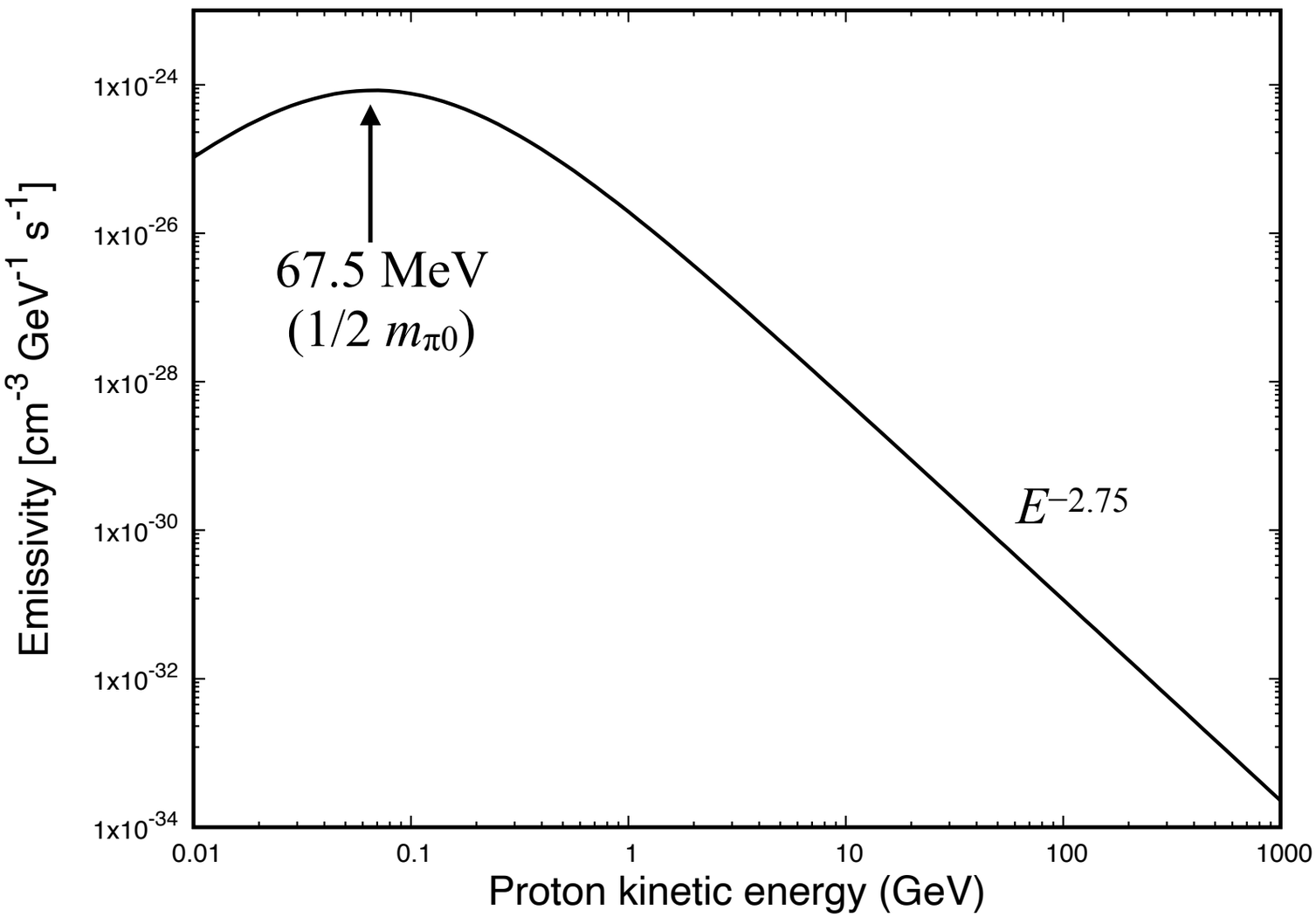
This can be an assignment question.

Thoudam, 2006, Astroparticle Physics 2006, 25, 328
Aharonian & Atoyan, Astronomy and Astrophyssics, 2000, 362, 937

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Gamma-ray emissivity: Emissivity from a proton power-law spectrum of $E^{-2.75}$ and a hydrogen density of $n_H \sim 1$ cm^{-3} .



The shape of the gamma-ray spectrum closely follow that of the proton spectrum at high energies.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Electrons/positrons emissivity:

$$p + p \rightarrow \pi^+$$
$$\hookrightarrow \mu^+ + \nu_\mu$$
$$\hookrightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

- The positron production involves a **three-step** process.
- Emissivity calculation will involve **three** integrals:
  - (1) Over the cosmic-ray kinetic energy.
  - (2) Over the π^0 total energy.
  - (3) Over the μ^+ total energy.

For details on the kinematics involved in the calculation, I recommend :
Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 125-128).

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Electrons/positrons emissivity:

Positron production term

$$q(E_{e^+}) = 2\pi n \int_{E_{\mathrm{th}_p}}^{\infty} dE_p I(E_p) \int_{M_e}^{E^*} \frac{dE'_e D(E'_e)}{\sqrt{E'^2_e - M^2_e}} \int_{(M_\mu/M^2_\pi)E_\pi E'_\pi(1-\beta_\pi\beta'_\pi)}^{(M_\mu/M^2_\pi)E_\pi E'_\pi(1+\beta_\pi\beta'_\pi)} dE_\mu$$

$$\times \int_{(M^2_\pi/M^2_\mu)(\eta-\beta_\mu\rho)E_\mu}^{(M^2_\pi/M^2_\mu)(\eta+\beta_\mu\rho)E_\mu} dE_\pi \frac{\sigma(E_\pi|E_p)}{(E^2_\pi - M^2_\pi)^{1/2}}$$

where,

$$D(E'_e) = \frac{4E'_e}{E^{*4}} (E'^2_e - M^2_e)^{1/2} \left[3(E^* - E'_e) + \frac{2}{3}\omega(4E'_e - 3E^*) \right]$$

$E^* = \frac{M_\mu}{2}\left[1 + \frac{M^2_e}{M^2_\mu}\right] \simeq \frac{M_\mu}{2}$ is the maximum energy available to the electron in the decay

$\omega \simeq 3/4$ is called Michel parameter

$I(E_\mathrm{p})$ is the cosmic-ray proton flux.

η and ρ are constants

β 's are relativistic velocities

$\sigma(E_\pi|E_p)$ is the differential cross-section

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 128).

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Neutrinos emissivity:

$$p + p \rightarrow \pi^+$$
$$\hookrightarrow \mu^+ + \nu_\mu \ \checkmark$$
$$\hookrightarrow e^+ + \nu_e + \bar{\nu}_\mu \ \checkmark$$

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Neutrinos emissivity:

$$p + p \rightarrow \pi^+$$
$$\hookrightarrow \mu^+ + \nu_\mu \ \checkmark$$
$$\hookrightarrow e^+ + \nu_e + \bar{\nu}_\mu \ \checkmark$$

- Emissivity calculation is little bit more complicated.

- Two types of neutrinos involved (ν_μ and ν_e): No. of ν_e is a factor 2 less than ν_μ .

A detailed calculation is given in Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Neutrinos emissivity:

$$p + p \rightarrow \pi^+$$
$$\hookrightarrow \mu^+ + \nu_\mu \ \checkmark$$
$$\hookrightarrow e^+ + \nu_e + \bar{\nu}_\mu \ \checkmark$$

- Emissivity calculation is little bit more complicated.

- Two types of neutrinos involved (ν_μ and ν_e): No. of ν_e is a factor 2 less than ν_μ .

- Muon neutrinos (ν_μ) are produced from two processes:
  (1) $\nu_{\mu 1}$ from the direct decay of π^+: **Two-step** process.
  (2) $\nu_{\mu 2}$ from the decay of μ^+: **Three-step** process.

A detailed calculation is given in Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Neutrinos emissivity:

$$p + p \rightarrow \pi^+$$
$$\hookrightarrow \mu^+ + \nu_\mu \ \checkmark$$
$$\hookrightarrow e^+ + \nu_e + \bar{\nu}_\mu \ \checkmark$$

- Emissivity calculation is little bit more complicated.

- Two types of neutrinos involved (ν_μ and ν_e): No. of ν_e is a factor 2 less than ν_μ .

- Muon neutrinos (ν_μ) are produced from two processes:
  (1) $\nu_{\mu 1}$ from the direct decay of π^+: **Two-step** process.
  (2) $\nu_{\mu 2}$ from the decay of μ^+: **Three-step** process.

- Electron neutrinos (ν_e) are produced only from one process:
  (1) Decay from μ^+: **Three-step** process.

- Total neutrinos produced is the sum of all these components:

$$q_\nu = q_\nu(\mu_1) + q_\nu(\mu_2) + q_\nu(e) \simeq q_\nu(\mu_1) + q_\nu(\mu_2)$$

A detailed calculation is given in Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Neutrinos emissivity:

$$q_\nu(E_\nu) = cn_H \int_{E_\nu}^{\infty} \sigma_{\text{inel}}(E_p) J_p(E_p) F_v\left(x \ , E_p \right) \frac{dE_p}{E_p}$$

*For the component from $\pi^+ \rightarrow \mu^+ + \nu_\mu$

$$F_{\nu_\mu^{(1)}}(x, E_p) = B' \frac{\ln(y)}{y} \left(\frac{1 - y^{\beta'}}{1 + k' y^{\beta'} (1 - y^{\beta'})} \right)^4 \left[\frac{1}{\ln(y)} - \frac{4\beta' y^{\beta'}}{1 - y^{\beta'}} - \frac{4k'\beta' y^{\beta'} (1 - 2y^{\beta'})}{1 + k' y^{\beta'} (1 - y^{\beta'})} \right],$$

where $x = E_{\nu_\mu}/E_p$, $y = x/0.427$,

$$B' = 1.75 + 0.204L + 0.010L^2,$$

$$\beta' = \frac{1}{1.67 + 0.111L + 0.0038L^2},$$

$$k' = 1.07 - 0.086L + 0.002L^2.$$

Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Neutrinos emissivity:

$$q_\nu(E_\nu) = cn_H \int_{E_\nu}^{\infty} \sigma_{\rm inel}(E_p) J_p(E_p) F_v\Big(x \ , E_p \Big) \frac{dE_p}{E_p}$$

*For the component from $\pi^+ \rightarrow \mu^+ + \nu_\mu$

$$F_{\nu_\mu^{(1)}}(x, E_p) = B' \frac{\ln(y)}{y} \left(\frac{1 - y^{\beta'}}{1 + k' y^{\beta'} (1 - y^{\beta'})} \right)^4 \left[\frac{1}{\ln(y)} - \frac{4\beta' y^{\beta'}}{1 - y^{\beta'}} - \frac{4k'\beta' y^{\beta'} (1 - 2y^{\beta'})}{1 + k' y^{\beta'} (1 - y^{\beta'})} \right],$$

where $x = E_{\nu_\mu}/E_p$, $y = x/0.427$,

$$B' = 1.75 + 0.204L + 0.010L^2,$$

$$\beta' = \frac{1}{1.67 + 0.111L + 0.0038L^2},$$

$$k' = 1.07 - 0.086L + 0.002L^2.$$

*For the component from $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$

$F_{\nu_\mu^{(2)}}(x, E_p)$

or

$$F_e(x, E_p) = B_e \frac{(1 + k_e(\ln x)^2)^3}{x(1 + 0.3/x^{\beta_e})} (-\ln(x))^5,$$

$$B_e = \frac{1}{69.5 + 2.65L + 0.3L^2},$$

$$\beta_e = \frac{1}{(0.201 + 0.062L + 0.00042L^2)^{1/4}},$$

$$k_e = \frac{0.279 + 0.141L + 0.0172L^2}{0.3 + (2.3 + L)^2}.$$

Here $x = E_e/E_\pi$ and $L = \ln(E_p/1\ \text{TeV})$.

Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Similarly, gamma-ray emissivity from Kelner et al. (2006):

$$q_\gamma(E_\gamma) = cn_H \int_{E_\gamma}^{\infty} \sigma_{\rm inel}(E_p) J_p(E_p) F_\gamma\left(x \ , E_p \right) \frac{dE_p}{E_p}$$

$$p + p \rightarrow \pi^0 \rightarrow \gamma + \gamma$$

where $F_\gamma(x, E_p) = B_\gamma \dfrac{d}{dx}\left[\ln(x) \left(\dfrac{1 - x^{\beta_\gamma}}{1 + k_\gamma x^{\beta_\gamma}(1 - x^{\beta_\gamma})} \right)^4 \right]$

$$= B_\gamma \frac{\ln(x)}{x} \left(\frac{1 - x^{\beta_\gamma}}{1 + k_\gamma x^{\beta_\gamma}(1 - x^{\beta_\gamma})} \right)^4 \left[\frac{1}{\ln(x)} - \frac{4\beta_\gamma x^{\beta_\gamma}}{1 - x^{\beta_\gamma}} - \frac{4k_\gamma \beta_\gamma x^{\beta_\gamma}(1 - 2x^{\beta_\gamma})}{1 + k_\gamma x^{\beta_\gamma}(1 - x^{\beta_\gamma})} \right],$$

$$x = E_\gamma / E_p$$

$$B_\gamma = 1.30 + 0.14L + 0.011L^2,$$

$$\beta_\gamma = \frac{1}{1.79 + 0.11L + 0.008L^2},$$

$$k_\gamma = \frac{1}{0.801 + 0.049L + 0.014L^2},$$

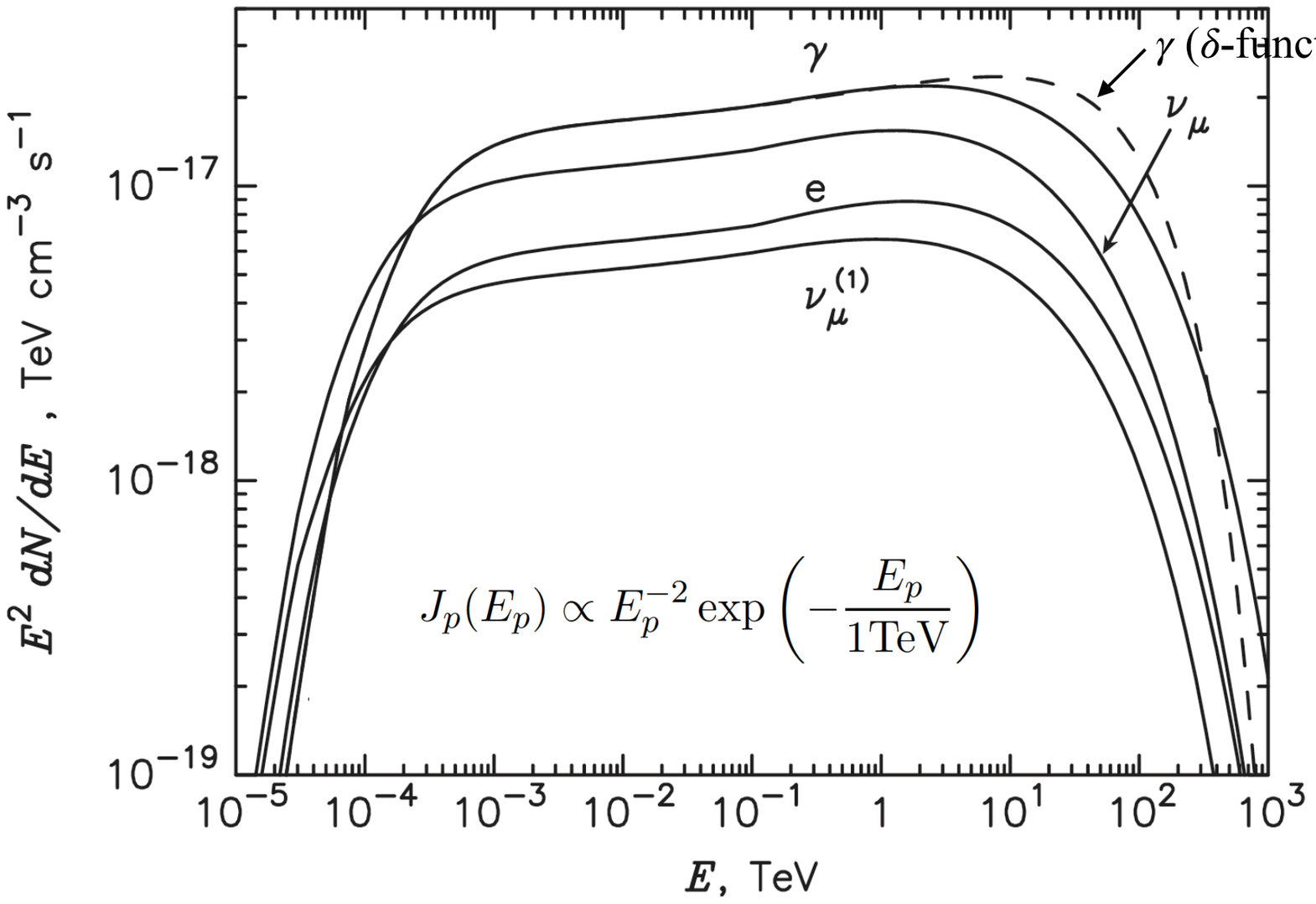
$$L = \ln(E_p / 1 \text{ TeV}).$$

Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Proton-proton inelastic collision

Emissivities of gammas, electrons and neutrinos from Kelner+ (2006) for a proton spectrum following a power-law spectrum with an exponential cut-off.



A factor of ~ 1.5 is usually multiplied to the emissivities to account for extra contributions from p-α, α-p, α-α interactions.

Kelner et al. Physical Review D, 2006, 74, 034018.

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(2) Proton-photon inelastic collision: Mostly relevant for extra-galactic sources

e± pair creation: $p + \gamma \rightarrow p + e^+ e^-$

- Threshold energy ~ 7x10^{17} eV on CMB
- Secondaries: Electron and positron

Photopion production: $p + \gamma \rightarrow p + \pi^0 \rightarrow p + \gamma\gamma$, [1/3 of all cases]

$\rightarrow n + \pi^+ \rightarrow n + \mu^+ + \nu_\mu$ [2/3 of all cases]

$\hookrightarrow e^+ \bar{\nu}_\mu \nu_e .$

- Threshold energy ~ 10^{20} eV on CMB
- Secondaries: Gammas, electrons, positrons, and neutrinos (ν_μ and ν_e)

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(2) Proton-photon inelastic collision:

e± pair creation: $p + \gamma \rightarrow p + e^+ e^-$

- Threshold energy ~ 7x10^{17} eV on CMB
- Secondaries: Electron and positron

Photopion production: $p + \gamma \rightarrow p + \pi^0 \rightarrow p + \gamma\gamma$, [1/3 of all cases]

$\rightarrow n + \pi^+ \rightarrow n + \mu^+ + \nu_\mu$ [2/3 of all cases]

$\hookrightarrow e^+ \bar{\nu}_\mu \nu_e .$

- Threshold energy ~ 10^{20} eV on CMB
- Secondaries: Gammas, electrons, positrons, and neutrinos (ν_μ and ν_e)

These threshold calculations requires that the electron-positron pair production and the photopion production processes take place when the energy of the photon in the rest frame of the projectile proton exceeds respectively:

$$2m_e c^2 \simeq 1\,\mathrm{MeV} \text{ and } m_\pi c^2 (1 + m_\pi / 2m_p) \simeq 145\mathrm{MeV}$$

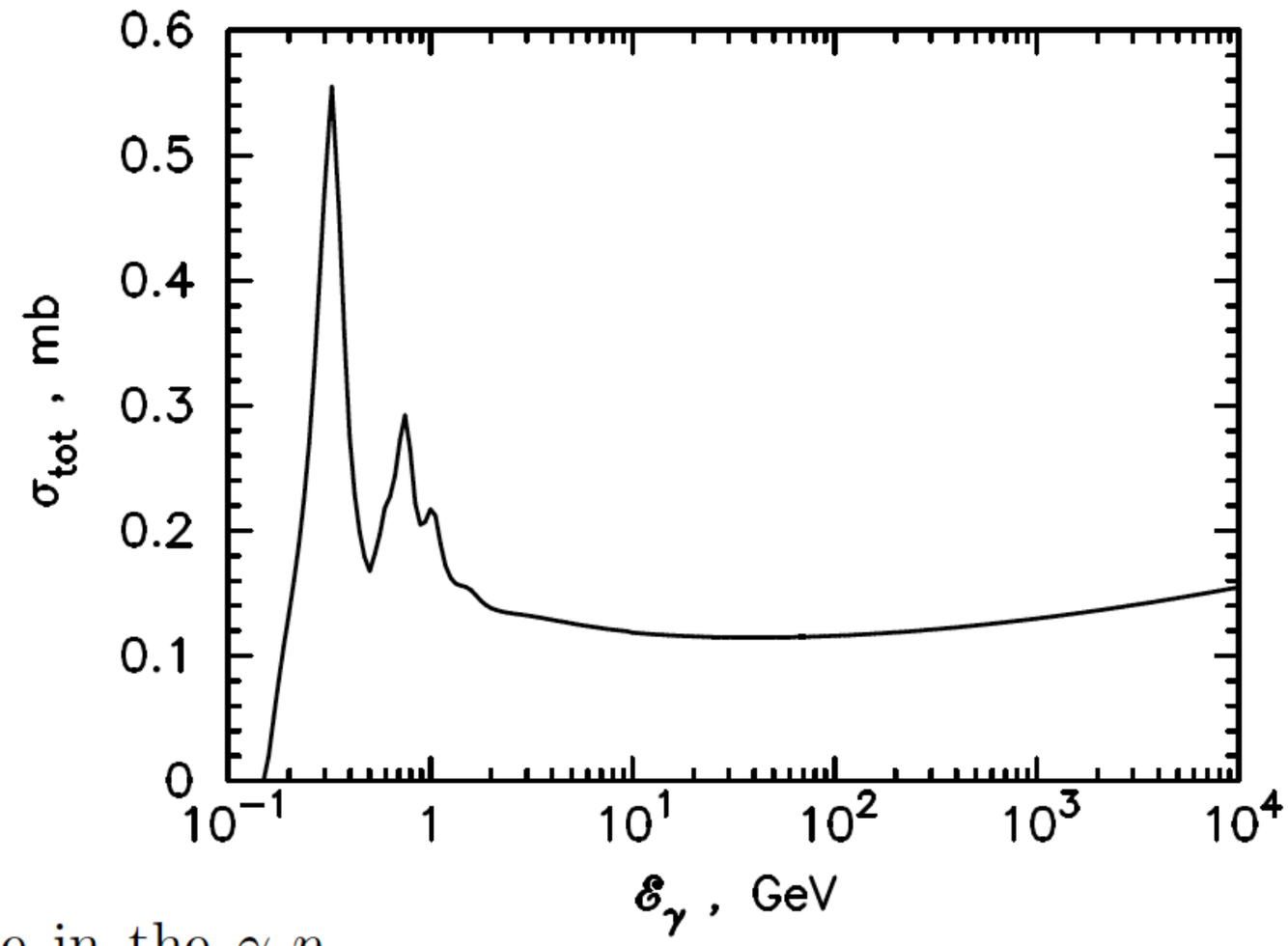
S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(2) Proton-photon inelastic collision:

Cross-section of p-γ interaction as a function of photon energy in the proton rest frame. Calculation performed using the SOPHIA code.



(Photon energy in proton rest frame)

Although the production of pions dominate in the γ-p interactions, some other channels, in particular the ones leading to production of K- and η-mesons, contribute noticeably (up to 10 to 20 %) to the overall production of photons and leptons.

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(2) Proton-photon inelastic collision:

The cross-section of (e^+, e^-) pair production (often called as Bethe-Heitler cross-section) is quite large, but only a small ($\leq 2m_e/m_p$) fraction of the proton energy is converted to the secondary electrons. The cross-section of photomeson production is smaller, but instead a substantial (10 percent or more) fraction of the proton energy is transferred to the secondary product. As a result, when the proton energy exceeds the π-meson production threshold, the hadronic interactions of protons dominate over the pair-production.

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(2) Proton-photon inelastic collision:

The energy losses of protons in the photon fields, in particular in the context of interactions of highest energy cosmic rays with 2.7 K cosmic microwave background radiation (CMBR), have been comprehensively studied

Less attention has been given to the calculations of the energy distribution of the products.

The cross-sections of secondary electrons and gamma-rays with the ambient photons significantly exceeds the cross-sections of interactions of protons with the same target photons. Therefore the electrons and gamma-rays cannot leave the active regions of their production, but rather trigger electromagnetic cascades in the surrounding radiation and magnetic fields. The spectra of gamma-rays formed during the cascade development are not sensitive to the initial energy distributions

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

(2) Proton-photon inelastic collision:

However, neutrinos which freely escape the source and thus have a undistorted imprint of parent protons.

And, at some specific conditions, the secondary electrons from the pair- and photomeson production processes can cool mainly through synchrotron radiation for which the source can be optically thin (see e.g. [7, 8, 9, 10]). Thus the exact calculations of spectra of secondary electrons are quite important, since the synchrotron radiation of these electrons carry direct information about the energy spectra of accelerated protons.

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

The production rate per unit energy interval of a daughter particle "*b*" (e.g. meson) from a parent "*a*" (e.g. proton) is given by:

$$Q_b(E_b) = \sum_{\mathrm{IT}} \int dE_a \, N_a(E_a) \, \Gamma^{\mathrm{IT}}_{a\to b}(E_a) \, \frac{dn^{\mathrm{IT}}_{a\to b}}{dE_b}(E_a, E_b) \, .$$

Here, $N_a(E_a)$ is the differential spectrum of parent particles[1] (particles per energy interval), and $\Gamma^{\mathrm{IT}}_{a\to b}$ is the interaction or decay rate (probability per unit time and particle) for the process $a \to b$ as a function of energy E_a (which is assumed to be zero below the threshold). Since in pion photoproduction many interaction types contribute, we split the production probability in interaction types "IT."

The function $dn^{\mathrm{IT}}_{a\to b}/dE_b(E_a, E_b)$ describes the distribution (as a function of parent and daughter energy) of daughter particles of type b per final state energy interval dE_b. This function can be non-trivial. It contains the kinematics of the decay process,

Hümmer, S. et al., The Astrophysical Journal, 2010, 721, 630

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

Interaction rate of the proton for an isotropic photon distribution:

$$\Gamma^{\mathrm{IT}}_{p\,\gamma\to p'\,b}(E_p) = \frac{1}{2}\frac{m_p^2}{E_p^2}\int_{\frac{\epsilon_{\mathrm{th}} m_p}{2E_p}}^{\infty} d\varepsilon \frac{n_\gamma(\varepsilon)}{\varepsilon^2}\int_{\epsilon_{\mathrm{th}}}^{2E_p\varepsilon/m_p} d\epsilon_r\, \epsilon_r\, \sigma^{\mathrm{IT}}(\epsilon_r)$$

where $n_\gamma(\varepsilon)$ is the photon density as a function of photon energy ε

$\sigma^{\mathrm{IT}}(\epsilon_r)$ is the proton-γ interaction cross-section,

$\epsilon_r = E_p\varepsilon/m_p(1-\cos_{p\gamma})$ is the photon energy in the proton rest frame

Here, $\epsilon_{\mathrm{th}} \simeq 150\,\mathrm{MeV}$ is the threshold below which the cross sections are zero.

In the equation, the integral over ϵ_r takes into account that the proton and photon may hit each other in different directions. In the most optimistic case, $\epsilon_r \approx 2E_p\varepsilon/m_p$ (head-on hit), whereas in the most pessimistic case, $\epsilon_r \approx 0$ (photon and proton travel in the same direction).

Hümmer, S. et al., The Astrophysical Journal, 2010, 721, 630

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

Energy distribution function of the daughter particle in a delta-function approximation:

$$\frac{dn_{a\to b}^{\mathrm{IT}}}{dE_b}(E_a, E_b) \simeq \delta\left(E_b - \chi_{a\to b}^{\mathrm{IT}} E_a\right) \cdot M_b^{\mathrm{IT}}$$

$\chi_{a\to b}^{\mathrm{IT}}$ describes the mean fraction of parent energy transferred into a daughter species. This is related to the inelasticity K (fractional energy loss of a proton) as: K= Sum of all $\chi_{a\to b}^{\mathrm{IT}}$ over all daughter particles.

M_b^{IT} describes the number of daughter particles produced at energy E_b

Hümmer, S. et al., The Astrophysical Journal, 2010, 721, 630

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

Using the functions $\Gamma^{\rm IT}_{p\,\gamma\to p'\,b}(E_p)$ and $\dfrac{dn^{\rm IT}_{a\to b}}{dE_b}(E_a, E_b)$ described in the previous slides, the spectrum of the daughter particles is calculated as:

$$Q_b(E_b) = \int_{E_b}^{\infty} \frac{dE_p}{E_p}\, N_p(E_p) \int_{\frac{\epsilon_{\rm th} m_p}{2E_p}}^{\infty} d\varepsilon\, n_\gamma(\varepsilon) R_b(x, y),$$

where $x \equiv \dfrac{E_b}{E_p}$, $\quad y \equiv \dfrac{E_p\,\varepsilon}{m_p}$,

$$R_b(x, y) \equiv \sum_{\rm IT} R^{\rm IT}(x, y) \equiv \sum_{\rm IT} \frac{1}{2y^2} \int_{\epsilon_{\rm th}}^{2y} d\epsilon_r\, \epsilon_r\, \sigma^{\rm IT}(\epsilon_r) M_b^{\rm IT}(\epsilon_r)\, \delta(x - \chi^{\rm IT}(\epsilon_r)).$$

which describes the energy distribution of the daughter particle over all interaction channels.

$\epsilon_{\rm th} \lesssim 2y = 2E_p\varepsilon/m_p \lesssim 10^4\,{\rm GeV}$

$\chi^{\rm IT}_{a\to b}$: Mean fraction of parent energy transferred into a daughter species.

$M_b^{\rm IT}$: Number of daughter particles produced at energy E_b

Hümmer, S. et al., The Astrophysical Journal, 2010, 721, 630

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

Using the functions $\Gamma^{\mathrm{IT}}_{p\,\gamma\to p'\,b}(E_p)$ and $\frac{dn^{\mathrm{IT}}_{a\to b}}{dE_b}(E_a, E_b)$ described in the previous slides, the spectrum of the daughter particles is calculated as:

$$Q_b(E_b) = \int_{E_b}^{\infty} \frac{dE_p}{E_p} N_p(E_p) \int_{\frac{\epsilon_{\mathrm{th}} m_p}{2E_p}}^{\infty} d\varepsilon\, n_\gamma(\varepsilon) R_b(x, y),$$

$n_\gamma(\varepsilon)$ is the target photon spectrum.

$N_p(E_p)$ is the incident proton spectrum.

* If the response function $R_b(x,y)$ is known, the secondary spectra can be calculated for any arbitrary injection proton and photon spectra.

* Using the equation above, the spectra of π^0, $\pi^{\pm}$ can be calculated.

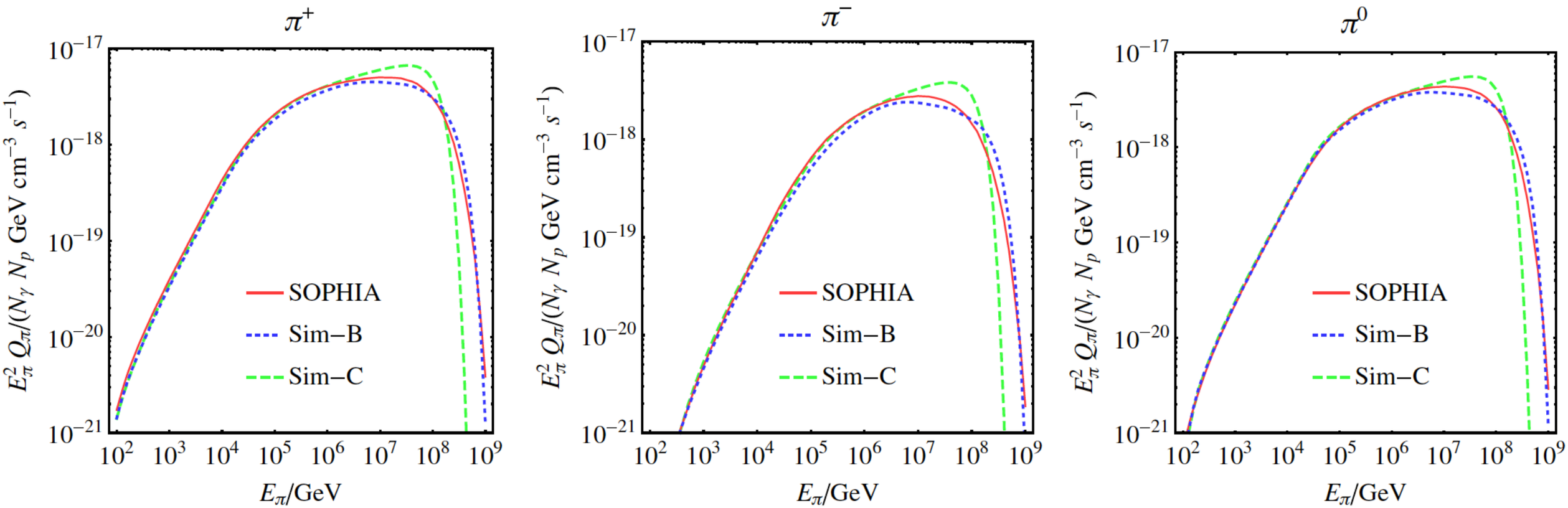
Hümmer, S. et al., The Astrophysical Journal, 2010, 721, 630

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

Pion spectra produced inside GRBs



Photon spectrum: $n_\gamma(\epsilon) = \begin{cases} n_0 \left(\frac{\varepsilon}{\text{GeV}}\right)^{-1} \frac{1}{\text{GeV cm}^3} & 0.2\,\text{eV} \leqslant \varepsilon \leqslant 1\,\text{keV} \\ n_0\, 10^{-6} \left(\frac{\varepsilon}{\text{GeV}}\right)^{-2} \frac{1}{\text{GeV cm}^3} & 1\,\text{keV} \leqslant \varepsilon \leqslant 300\,\text{keV} \end{cases}$

Proton spectrum: $N_p(E_p) = N_0 \left(\frac{E_p}{\text{GeV}}\right)^{-2} \exp\left[-\left(\frac{E_p}{6.9 \cdot 10^8\,\text{GeV}}\right)^2\right] \times \frac{1}{\text{GeV cm}^3} \quad E_p \geqslant 1\,\text{GeV}.$

Hümmer, S. et al., The Astrophysical Journal, 2010, 721, 630

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

Photopion production:

* Once the pion spectra are obtained, the spectra of the decay products can be calculated using a similar procedure described for the proton-proton interaction.

$$p + p \rightarrow \pi^0 \rightarrow \gamma + \gamma$$

Products: γ_s

$$p + p \rightarrow \pi^+ \rightarrow \mu^+ + \nu_\mu \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

Products: $e^{\pm}$, $\bar{\nu}_e$, $\bar{\nu}_\mu$, ν_μ and ν_e

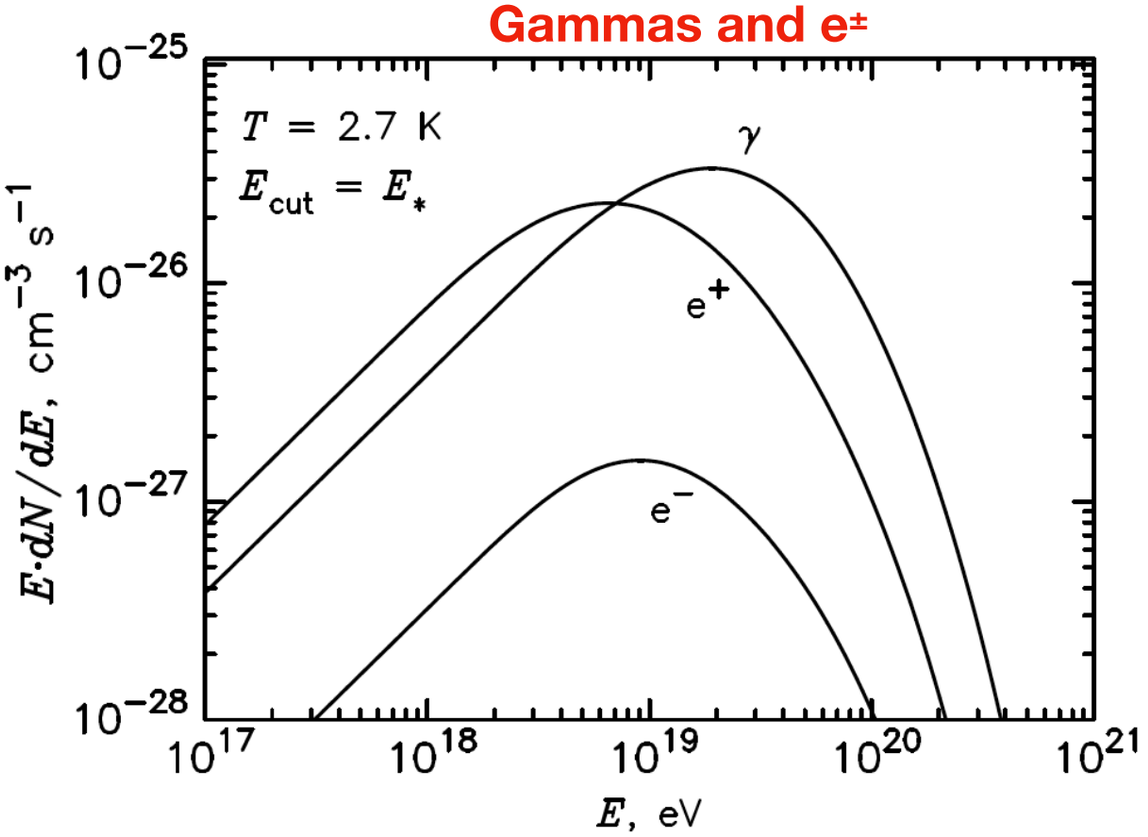
Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

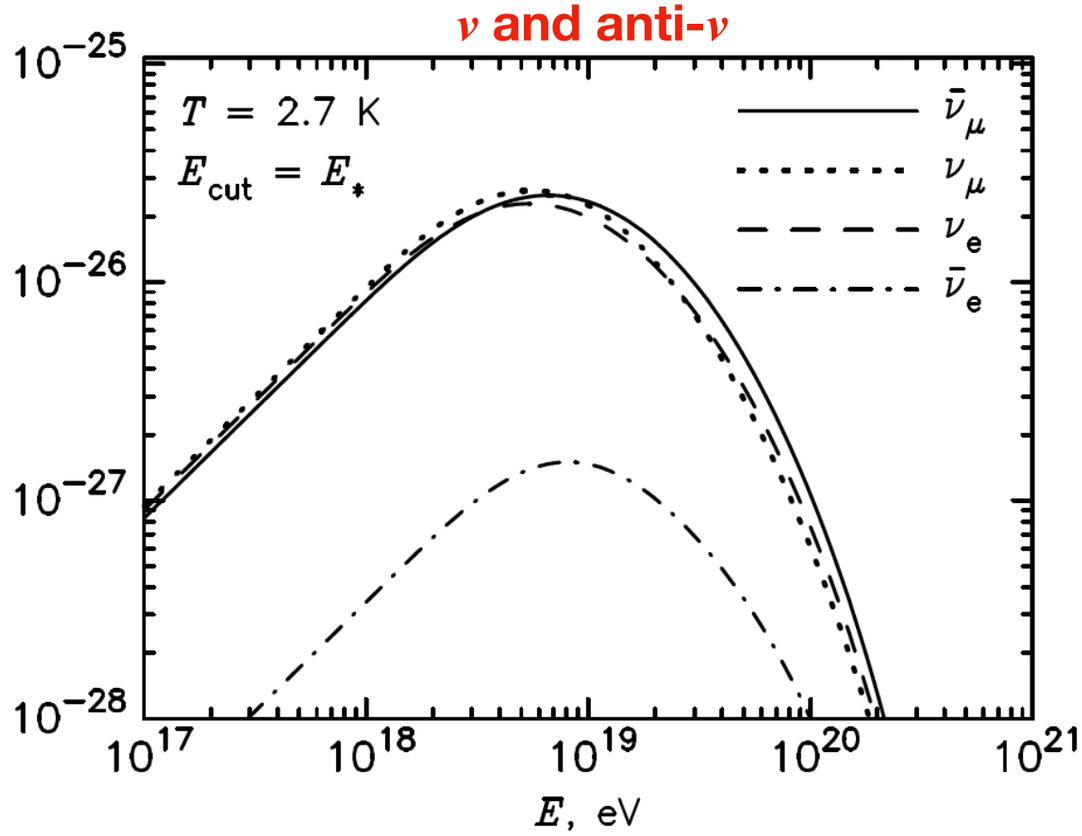
Photopion production:

Production spectra of decay products from photopion production

Gammas and e±



ν and anti-*ν*



Proton spectrum: $N_p(E_p) = AE_p^{-2} \exp\left(-\frac{E_p}{E_{\rm cut}}\right)$ $\qquad E_{\rm cut} \approx 3.0 \cdot 10^{20}$ eV

Photon: Cosmic microwave background

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

e± pair production: $p + \gamma \rightarrow p + e^+ e^-$

Spectrum of electrons/positrons from a power-law spectrum of proton:

$$\frac{dN}{dE_e} = \int_{\gamma_{\rm th}}^{\infty} N_p(\gamma_{\rm p})\, d\gamma_p \; \frac{1}{2\gamma_p^3} \int_{\frac{(\gamma_p+E_e)^2}{4\gamma_p^2 E_e}}^{\infty} d\epsilon \, \frac{f_{\rm ph}(\epsilon)}{\epsilon^2} \int_{\frac{(\gamma_p+E_e)^2}{2\gamma_p E_e}}^{2\gamma_p\epsilon} d\omega\, \omega \times \int_{\frac{\gamma_p^2+E_e^2}{2\gamma_p E_e}}^{\omega-1} \frac{dE_-}{p_-} W(\omega, E_-, \xi)$$

$\gamma_{\rm p}$ is the proton Lorentz factor.

$\gamma_{\rm th} \sim m_{\rm e}/\epsilon$ is the proton threshold Lorentz factor, ϵ is the photon energy.

$f_{\rm ph}(\epsilon)$ is the photon distribution function.

E_- and P_- are the electron energy and momentum in the proton rest frame.

$W(\omega, E_-, \xi)$ is the double-differential cross-section as a function of energy and emission angle of the electron in the rest frame of the proton.

$\omega = \epsilon\gamma_p\,(1 - \cos\theta)$ is the energy of the photon the rest frame of proton.

$\xi = (\gamma_p E_- - E_e)/(\gamma_p p_-)$.

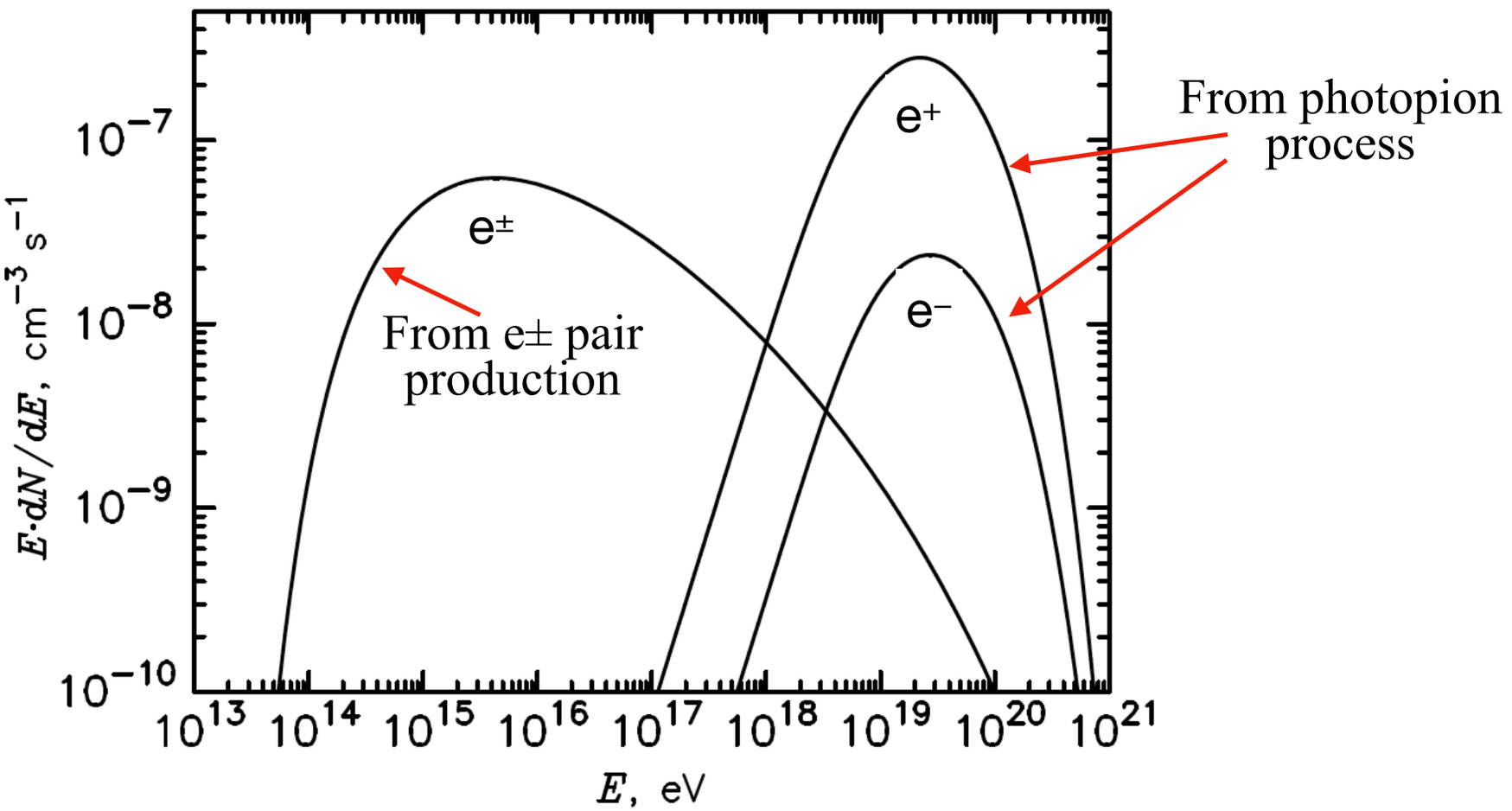
S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from hadronic processes

e± pair production:

Production spectra of electrons/positrons from pair production



Proton spectrum: $N_p(E_p) = AE_p^{-2} \exp\left(-\frac{E_p}{E_{\rm cut}}\right)$ $E_{\rm cut} \approx 3.0 \cdot 10^{20}$ eV

Photon: Cosmic microwave background

S. R. Kelner and F. A. Aharonian Phys. Rev. D, 2008, **78**, 034013

Emission of high-energy gamma rays, neutrinos and secondary $e^{\pm}$

Emission from leptonic processes

Electron-positron annihilation:

$$e^+ + e^- \longrightarrow \gamma + \gamma$$

A significant fraction of the leptonic component of high-energy particles could be in the form of positrons. There are several sources of positrons in astronomical environments. They can be produced from:

- Decay of π^+
- β decay of radioactive nuclei
- $e^{\pm}$ pair production from photo-matter photo-photon interaction

Emission of high-energy gamma rays, neutrinos and secondary $e^{\pm}$

Emission from leptonic processes

Electron-positron annihilation:

$$e^+ + e^- \longrightarrow \gamma + \gamma$$

- **Annihilation at rest:** Both photons have energies equal to 0.511 MeV (the rest-mass energy of an electron).

- **Annihilation in flight:** There is a dispersion in photon energies.

E.g: if the positron is moving with a velocity v and Lorentz factor γ, and the electron is at rest, the ejected photons have energies given by

$$E = \frac{m_e c^2 (1+\gamma)}{2}\left(1 \pm \frac{V}{c}\right)$$

\+ —> in the direction of the positron
− —> opposite

where $V = \gamma v (1+\gamma)$

High Energy Astrophysics, Longair: Page 277-278

Emission of high-energy gamma rays, neutrinos and secondary $e^{\pm}$

Emission from leptonic processes

Electron-positron annihilation:

$$e^+ + e^- \longrightarrow \gamma + \gamma$$

- **Annihilation at rest:** Both photons have energies equal to 0.511 MeV (the rest-mass energy of an electron).

- **Annihilation in flight:** There is a dispersion in photon energies.

E.g: if the positron is moving with a velocity v and Lorentz factor γ, and the electron is at rest, the ejected photons have energies given by

$$E = \frac{m_e c^2(1+\gamma)}{2}\left(1 \pm \frac{V}{c}\right)$$

\+ —> in the direction of the positron
− —> opposite

where $V = \gamma v(1+\gamma)$

- The cross-section in extreme relativistic limit is: $\sigma = \frac{\pi r_e^2}{\gamma}[\ln 2\gamma - 1]$

- At thermal energies: $\sigma \approx \frac{\pi r_e^2}{(v/c)}$ where $r_e = 2.8 \times 10^{15}$ m is the classical electron radius.

High Energy Astrophysics, Longair: Page 277-278

Emission of high-energy gamma rays, neutrinos and secondary $e^{\pm}$

Emission from leptonic processes

Electron-positron annihilation:

$$e^+ + e^- \longrightarrow \gamma + \gamma$$

- The gamma-ray production involves a **one-step** process.
- Emissivity calculation will involve **one** integral:
  (1) Over the incident positron kinetic energy.

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Electron-positron annihilation:

Gamma-ray emissivity from a given spectrum of positrons interacting with ambient electrons:

$$q(\varepsilon) = 4\pi n_e \int_{\gamma_{+\min}}^{\infty} I_{e+}(\gamma_+) d\gamma_+ \times \frac{d\sigma(\varepsilon, \gamma_+)}{d\varepsilon}$$

where, $\varepsilon = E/m_e c^2$ is the gamma-ray energy,

$\gamma_+ = E_+/m_e c^2$ is the positron Lorentz factor,

n_e is the ambient electron number density,

$I_{e+}(\gamma_+)$ is the positron spectrum,

$$\frac{d\sigma(\varepsilon, \gamma_+)}{d\varepsilon} = \frac{\pi r_e^2 c n_e}{\gamma_+ p_+} \left[\left(\frac{\varepsilon}{\gamma_+ + 1 - \varepsilon} + \frac{\gamma_+ + 1 - \varepsilon}{\varepsilon} \right) + 2 \left(\frac{1}{\varepsilon} + \frac{1}{\gamma_+ + 1 - \varepsilon} \right) - \left(\frac{1}{\varepsilon} + \frac{1}{\gamma_+ + 1 - \varepsilon} \right)^2 \right]$$

$$p_+ = \sqrt{\gamma_+^2 - 1}$$

$$\gamma_{+\min} = \frac{2\varepsilon^2 - 2\varepsilon + 1}{2\varepsilon - 1}$$ is the minimum positron Lorentz factor.

Aharonian & Atoyan, Astronomy and Astrophysics, 2000, 362, 937
Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 41-47)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Electron-positron annihilation:

Gamma-ray emissivity from a given spectrum of positrons interacting with ambient electrons:

$$q(\varepsilon) = 4\pi n_e \int_{\gamma_{+\min}}^{\infty} I_{e+}(\gamma_+) d\gamma_+ \times \frac{d\sigma(\varepsilon, \gamma_+)}{d\varepsilon}$$

where, $\varepsilon = E/m_e c^2$ is the gamma-ray energy,

$\gamma_+ = E_+/m_e c^2$ is the positron Lorentz factor,

n_e is the ambient electron number density,

$I_{e+}(\gamma_+)$ is the positron spectrum,

For the power-law spectrum of the positrons $I_{e+} \propto \gamma_+^{-\Gamma_e}$, the spectrum of annihilation radiation at high energies $E \gg m_e c^2$ has a power-law form

$$J_{\rm ann}(E) \propto E^{-(\Gamma_e+1)} \left[\ln(2E/m_e c^2) - 1\right] .$$

Thus, the spectrum of annihilation radiation is steeper than the spectrum of positrons by an index 1.

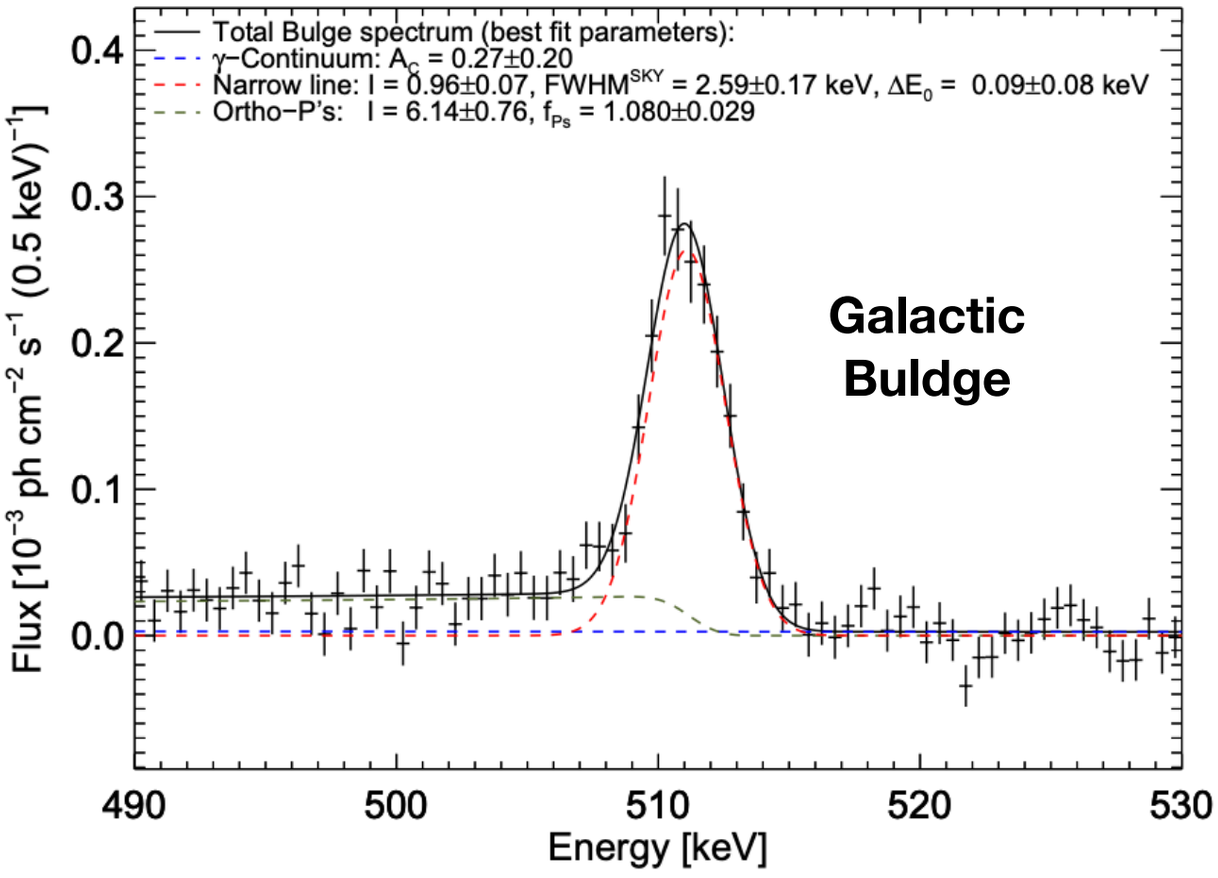
Aharonian & Atoyan, Astronomy and Astrophysics, 2000, 362, 937
Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 41-47)

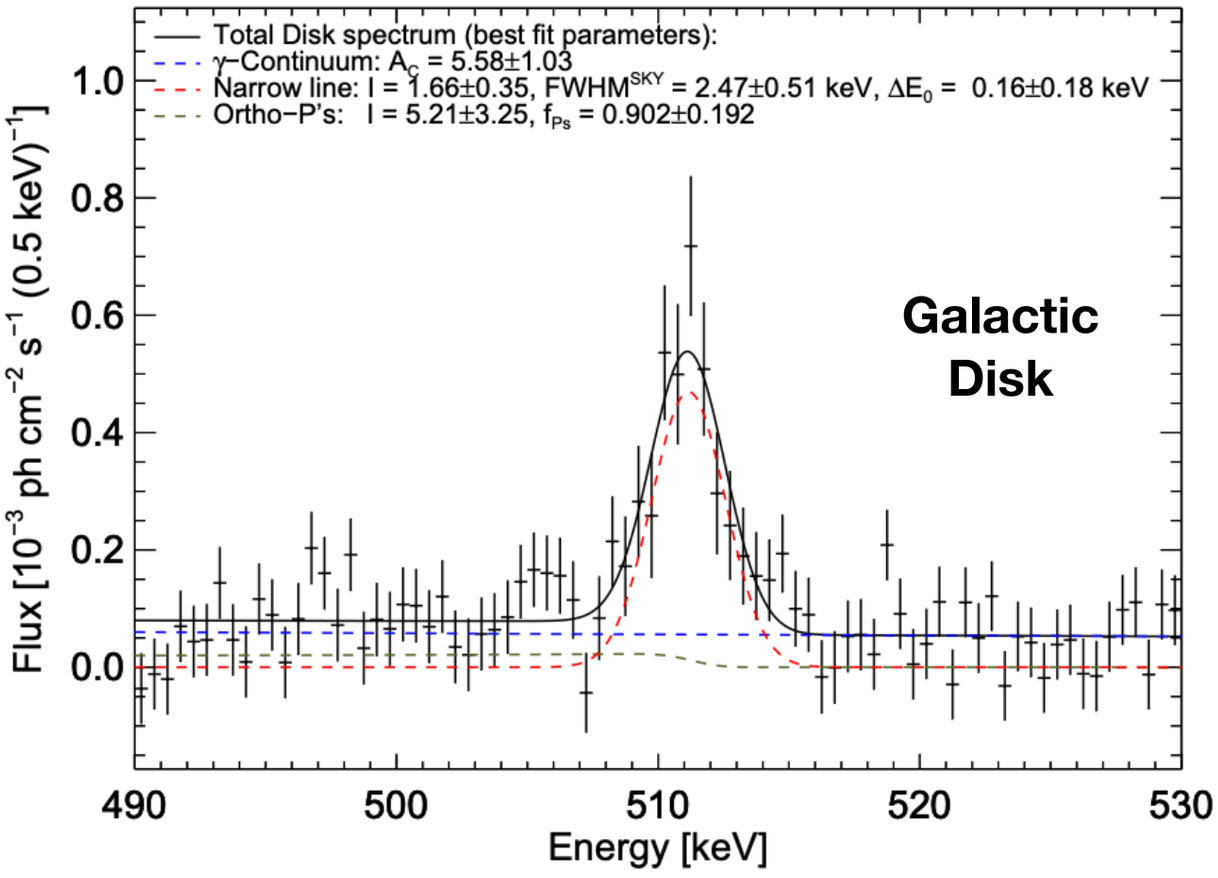
Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Electron-positron annihilation:

Spectrum of annihilation gamma-rays from the Galactic bulge and the disk





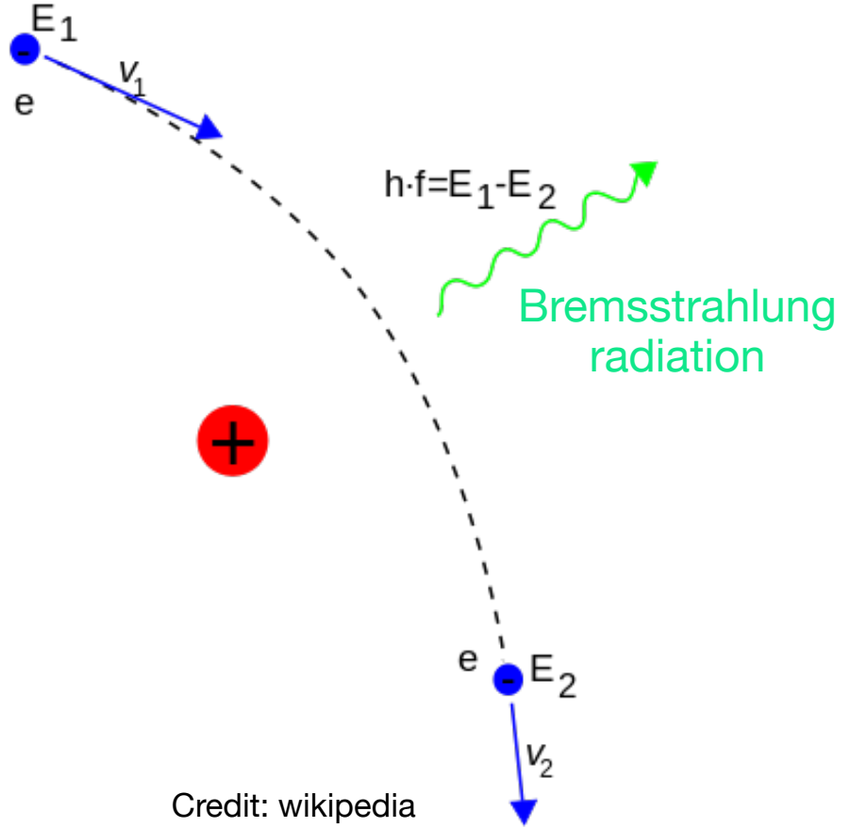
Siegert, T. et al, 2015, arXiv:1512.00325

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

Bremsstrahlung, which is a German word meaning "breaking radiation," is the radiation emitted by a charged particle accompanying deceleration. The cross sections for γ-ray production from bremsstrahlung are derived in Heitler's book (Heitler, 1960).



Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 153-155)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

Gamma-ray emissivity from bremsstrahlung emission of electrons in a medium:

$$q(E_\gamma) = 4\pi n_H \int_{E_\gamma}^{\infty} I_e(E) dE \times \frac{d\sigma(E_\gamma, E)}{dE_\gamma}$$

where, $E\gamma$ is the gamma-ray energy,

E is the electron energy,

n_{H} is the ambient atom density,

$I_e(E)$ is the electron spectrum,

$\frac{d\sigma(E_\gamma, E)}{dE_\gamma}$ is the differential cross-section for the production of gamma rays of energy $E\gamma$ by an electron of energy E.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 153-155)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

The differential cross-section can be written as:

$$\frac{d\sigma(E_\gamma, E)}{dE_\gamma} = \sigma_b(E, E_\gamma) f_b(E_\gamma|E)$$

where, $\sigma_b(E, E_\gamma)$ is the cross-section for bremsstrahlung.

$f_b(E_\gamma|E)$ is the normalized distribution of gamma rays.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 153-155)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

The differential cross-section can be written as:

$$\frac{d\sigma(E_\gamma, E)}{dE_\gamma} = \sigma_b(E, E_\gamma) f_b(E_\gamma|E)$$

where, $\sigma_b(E, E_\gamma)$ is the cross-section for bremsstrahlung.

$f_b(E_\gamma|E)$ is the normalized distribution of gamma rays.

For relativistic particles, the cross-section for bremsstrahlung: $\sigma_b(E, E_\gamma) \simeq \frac{\langle M \rangle}{\langle X \rangle} \left(\frac{E}{E_\gamma}\right)$

where $\langle M \rangle$ is the average mass of the target atoms in grams and $\langle X \rangle$ is the average radiation length for the gas in grams per square cm.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 153-155)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

The differential cross-section can be written as:

$$\frac{d\sigma(E_\gamma, E)}{dE_\gamma} = \sigma_b(E, E_\gamma) f_b(E_\gamma|E)$$

where, $\sigma_b(E, E_\gamma)$ is the cross-section for bremsstrahlung.

$f_b(E_\gamma|E)$ is the normalized distribution of gamma rays.

For relativistic particles, the cross-section for bremsstrahlung: $\sigma_b(E, E_\gamma) \simeq \frac{\langle M \rangle}{\langle X \rangle} \left(\frac{E}{E_\gamma}\right)$

where $\langle M \rangle$ is the average mass of the target atoms in grams and $\langle X \rangle$ is the average radiation length for the gas in grams per square cm.

To a good approximation, especially in the case of relativistic bremsstrahlung (Heitler, 1960), the normalized distribution of γ-rays produced may be taken to be a square distribution given by

$$f(E_\gamma|E) = \begin{cases} E^{-1} & \text{for } 0 \leqslant E_\gamma \leqslant E \\ 0 & \text{otherwise} \end{cases}$$

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 153-155)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

*Substituting the expressions for $\sigma_b(E, E_\gamma)$ and $f_b(E_\gamma|E)$, the gamma-ray emissivity can be calculated as:

$$q(E_\gamma) = \frac{\langle M \rangle}{\langle X \rangle} \left[\int_{E_\gamma}^{\infty} dE \; I_e(E) \right] E_\gamma^{-1}$$

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Bremsstrahlung emission:

*Substituting the expressions for $\sigma_b(E, E_\gamma)$ and $f_b(E_\gamma|E)$, the gamma-ray emissivity can be calculated as:

$$q(E_\gamma) = \frac{\langle M \rangle}{\langle X \rangle} \left[\int_{E_\gamma}^{\infty} dE \; I_e(E) \right] E_\gamma^{-1}$$

*For a power-law spectrum of electron $I_e \propto E^{-\Gamma_e}$,

the gamma-ray spectrum follows: $q \propto E_\gamma^{-\Gamma_e}$

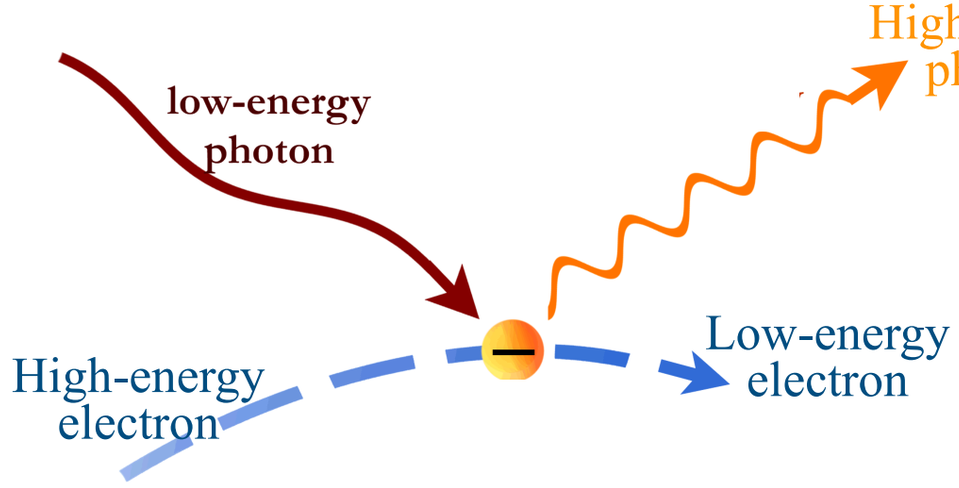
The gamma-ray spectral index remains the same as that of the electron spectrum.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 153-155)

Emission from leptonic processes

Inverse Compton emission:

Inverse Compton process: High-energy electron scatters a low-energy photon giving producing high-energy photon. The Electron losses energy during the process.



Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Inverse Compton emission:

Low-energy interaction cross-section

Under the condition $\gamma\epsilon \ll mc^2$, the differential cross section for the production of a γ-ray of energy $E_\gamma = \epsilon_1$ by Compton scattering is given by (Ginzburg and Syrovatskii, 1964)

$$\frac{d\sigma(E_\gamma|\epsilon,E)}{dE_\gamma} = \frac{\pi r_0^2}{4}\frac{(mc^2)^4}{\epsilon^2 E^3}\left\{2\frac{E_\gamma}{E} - \frac{(mc^2)^2 E_\gamma^2}{\epsilon E^3} + \frac{4E_\gamma}{E}\ln\frac{(mc^2)^2 E_\gamma}{4\epsilon E^2} + \frac{8\epsilon E}{(mc^2)^2}\right\}$$

and the mean energy of the γ-ray produced is given by

$$\langle E_\gamma\rangle = \frac{4}{3}\gamma^2\langle\epsilon\rangle$$

where, ϵ is the target photon energy,
γ is the electron Lorentz factor.
$E\gamma$ is the gamma-ray energy,
E is the electron energy,

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 149-153)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Inverse Compton emission:

Low-energy interaction cross-section

A good approximation is: $$\frac{d\sigma(E_\gamma|\epsilon,E)}{dE_\gamma} \simeq \sigma_T \delta\left(E_\gamma - \frac{4}{3}\,\langle\epsilon\rangle\,\frac{E^2}{(mc^2)^2}\right)$$

where, $\sigma_T = \frac{8}{3}\,\pi r_0^2 = 6.65 \times 10^{-25}\ \mathrm{cm}^2$ is the Thompson cross-section.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 149-153)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Inverse Compton emission:

High-energy interaction cross-section

At high energies, $\gamma\epsilon \gg mc^2$, Klein-Nishina cross-section works:

$$\sigma_c \rightarrow \pi r_0^2 \left(\frac{mc^2}{\epsilon}\right)\left[\frac{1}{2} + \ln\left(\frac{2\epsilon}{mc^2}\right)\right]$$

and the energy of the scattered photon is $\epsilon_1 \sim \gamma mc^2 \sim E$

(almost the same as the energy of the incident electron)

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 149-153)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Inverse Compton emission:

Gamma-ray emissivity from inverse compton process of electrons in a photon field:

$$q(E_\gamma) = 4\pi n_{ph} \int_{E_\gamma}^{\infty} I_e(E) dE \times \frac{d\sigma(E_\gamma|\epsilon, E)}{dE_\gamma}$$

where, $E\gamma$ is the gamma-ray energy,

E is the electron energy,

$n_{\rm ph}$ is the ambient photon density,

$I_e(E)$ is the electron spectrum,

$\frac{d\sigma(E_\gamma|\epsilon, E)}{dE_\gamma}$ is the differential cross-section for the production of gamma rays of energy $E\gamma$ by an electron of energy E.

ϵ is the target photon energy.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 149-153)

Emission of high-energy gamma rays, neutrinos and secondary e±

Emission from leptonic processes

Inverse Compton emission:

Taking, $$\frac{d\sigma(E_\gamma|\epsilon,E)}{dE_\gamma} \simeq \sigma_T \delta\left(E_\gamma - \frac{4}{3}\langle\epsilon\rangle \frac{E^2}{(mc^2)^2}\right)$$

The gamma-ray emissivity for a power-law electron spectrum $I_e(E) = K_e E^{-\Gamma}$ be obtained:

$$q(E_\gamma) = 4\pi n_{ph} \int_{E_\gamma}^{\infty} I_e(E)dE \times \frac{d\sigma(E_\gamma|\epsilon,E)}{dE_\gamma}$$

$$= n_{\rm ph}\sigma_T 4\pi \int dE\ K_e E^{-\Gamma} \delta\left(E_\gamma - \frac{4}{3}\langle\epsilon\rangle \frac{E^2}{(mc^2)^2}\right)$$

$$= n_{\rm ph}\sigma_T 4\pi K_e \left[\left(\frac{3}{4}\right)^{1/2} E_\gamma^{-1/2} \langle\epsilon\rangle^{-1/2} mc^2\right]^{-\Gamma} \frac{dE}{dE_\gamma}$$

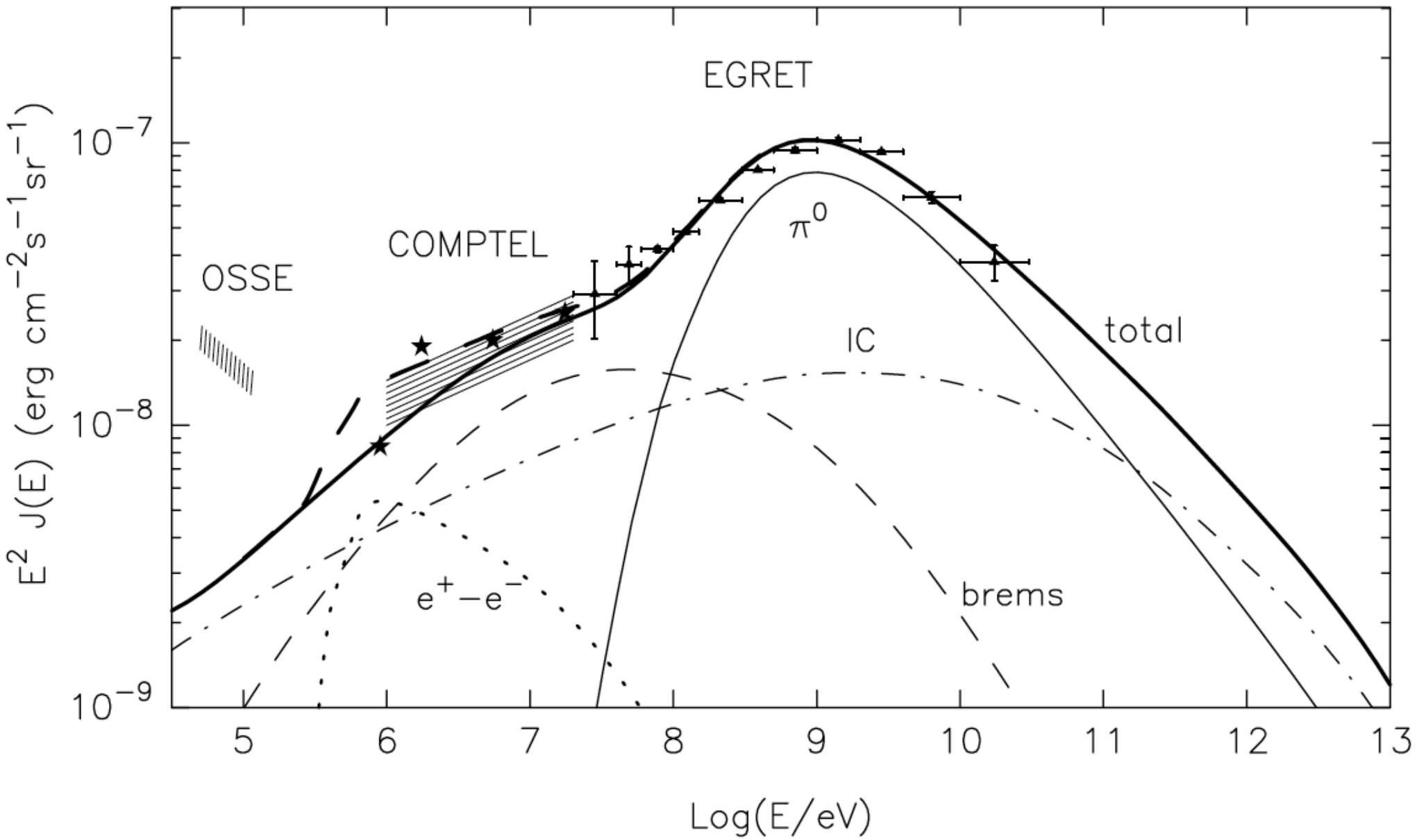
$$= \frac{1}{2}\, n_{\rm ph}\sigma_T 4\pi K_e (mc^2)^{1-\Gamma} \left(\frac{4}{3}\langle\epsilon\rangle\right)^{(\Gamma-1)/2} E_\gamma^{-(\Gamma+1)/2}$$

The gamma-ray spectrum is harder (smaller index) than the electron spectrum.

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 149-153)

Emission of high-energy gamma rays, neutrinos and secondary e±

Example: Diffuse gamma-ray flux observed from the Galactic disk

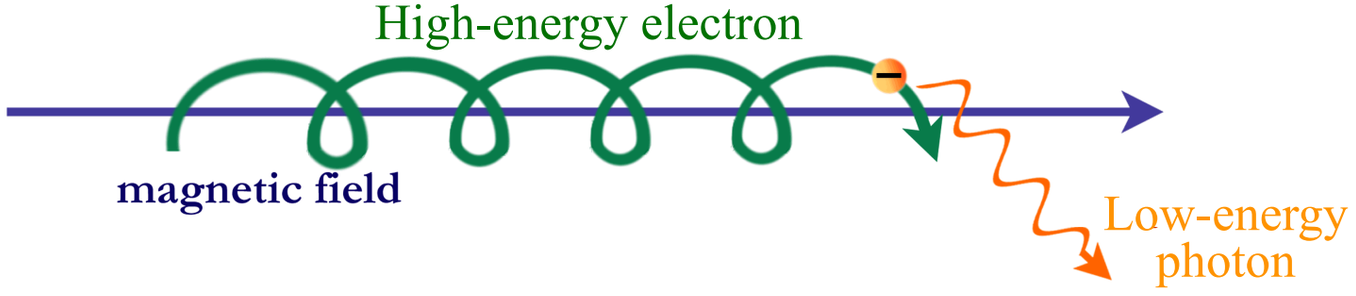


Note: The cosmic-ray proton and electron spectra used in the calculation are obtained after propagation through the Galaxy. They are different from that expected inside their sources.

Aharonian & Atoyan, Astronomy and Astrophysics, 2000, 362, 937

Emission of synchrotron radiation

Synchrotron process



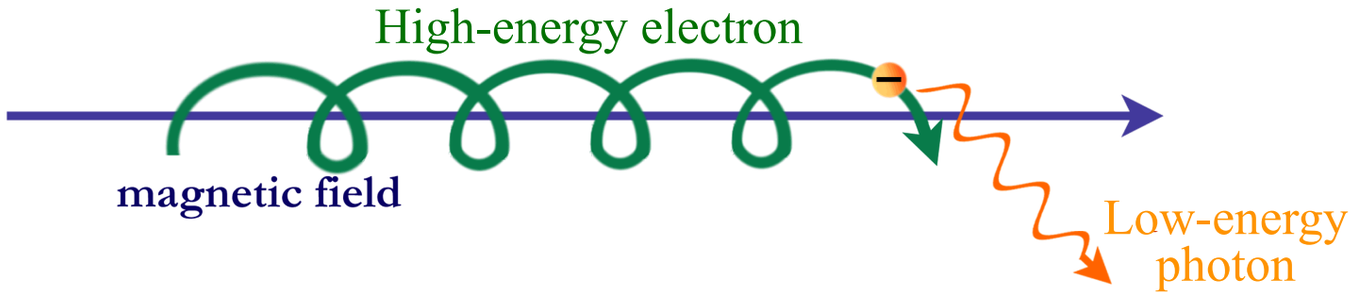
The synchrotron radiation, the emission of very relativistic and ultrarelativistic electrons gyrating in a magnetic field, is the process which dominates much of high energy astrophysics. It was originally observed in early betatron experiments in which electrons were first accelerated to ultrarelativistic energies. This process is responsible for the radio emission from the Galaxy, from supernova remnants and extragalactic radio sources. It is also responsible for the non-thermal optical and X-ray emission observed in the Crab Nebula and possibly for the optical and X-ray continuum emission of quasars.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Energy loss rate:



*The average energy loss rate for an electron of energy E due to synchrotron emission is given by,

$$-\left(\frac{\mathrm{d}E}{\mathrm{d}t}\right)_{\mathrm{sync}} = \frac{4}{3}\sigma_{\mathrm{T}}\, c u_{\mathrm{mag}} \left(\frac{v^2}{c^2}\right) \gamma^2$$

where $\sigma_T = 6.653 \times 10^{-29}$ m^2 is the Thompson crossection

$u_{mag} = B^2/2\mu_0$ is the energy density of the magnetic field

v: velocity of the incoming electron

γ: Lorentz factor of the incoming electron

*During its lifetime, it is likely that the high energy particle is randomly scattered in pitch angle.

Emission of synchrotron radiation

Synchrotron process

Energy loss rate:

$$-\left(\frac{\mathrm{d}E}{\mathrm{d}t}\right)_{\mathrm{sync}} = \frac{4}{3}\sigma_{\mathrm{T}}\, c u_{\mathrm{mag}} \left(\frac{v^2}{c^2}\right)\gamma^2 \qquad -\left(\frac{\mathrm{d}E}{\mathrm{d}t}\right)_{\mathrm{IC}} = \frac{4}{3}\sigma_{\mathrm{T}}\, c u_{\mathrm{rad}} \left(\frac{v^2}{c^2}\right)\gamma^2$$

*Notice the similarity to the energy loss rate by inverse Compton process.

This is not an accident. The reason for the similarity is that, in both cases, the electron is accelerated by the electric field which it observes in its instantaneous rest-frame. The electron does not really care about the origin of the electric field. In the case of synchrotron radiation, the constant accelerating electric field is associated with the motion of the electron through the magnetic field $\boldsymbol{B}$, $\boldsymbol{E'} = \boldsymbol{v} \times \boldsymbol{B}$, and, in the case of inverse Compton scattering, it is the sum of all the electric fields of the incident waves. Notice that, in the latter case, the fields of the waves add incoherently and it is the sum of the squares of the electric field strengths of the waves which appears in the formulae.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Emission pattern:

One of the general features of the radiation of relativistic electrons is that the radiation is *beamed* in the direction of motion of the electron. This is primarily associated with the effects of relativistic aberration between the instantaneous rest frame of the electron and the observer's frame of reference. In addition, we need to consider carefully the time development of the radiation detected by the distant observer.

Consider first an electron gyrating about the magnetic field direction at a pitch angle of 90°. The electron is accelerated towards its guiding centre, that is, radially inwards, and in its instantaneous rest frame emits dipole radiation with respect to the acceleration vector.

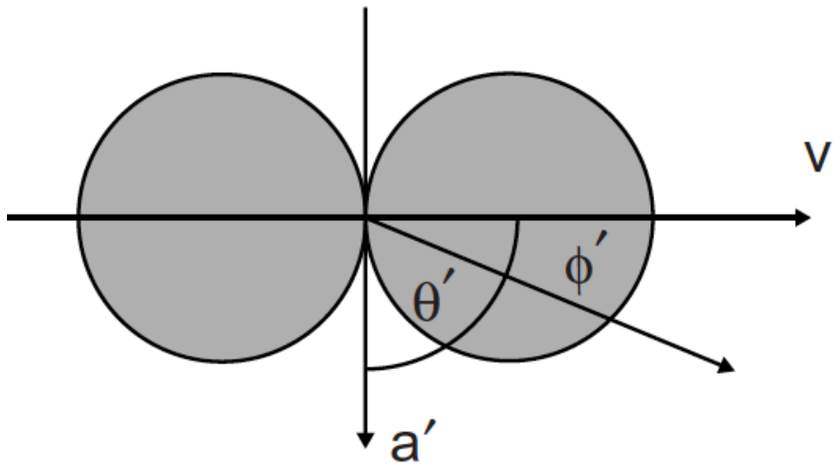
High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

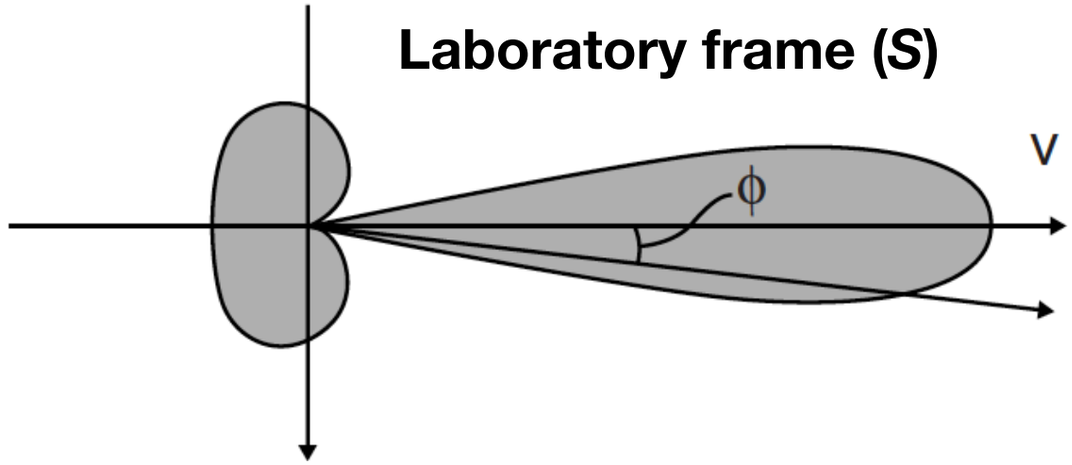
Emission pattern:

Electron's rest frame (*S'*)



To centre of particle's orbit

Laboratory frame (*S*)



To centre of particle's orbit

We can therefore work out the radiation pattern in the laboratory frame of reference by applying the aberration formulae with the results illustrated schematically in the diagrams. The angular distribution of radiation with respect to the velocity vector in the frame S′ is $I_\nu \propto \sin^2\theta' = \cos^2\phi'$. We may think of this as being the probability distribution with which photons are emitted by the electron in its rest frame. The appropriate aberration formulae between the two frames are:

$$\sin\phi = \frac{1}{\gamma}\frac{\sin\phi'}{1 + (v/c)\cos\phi'}$$

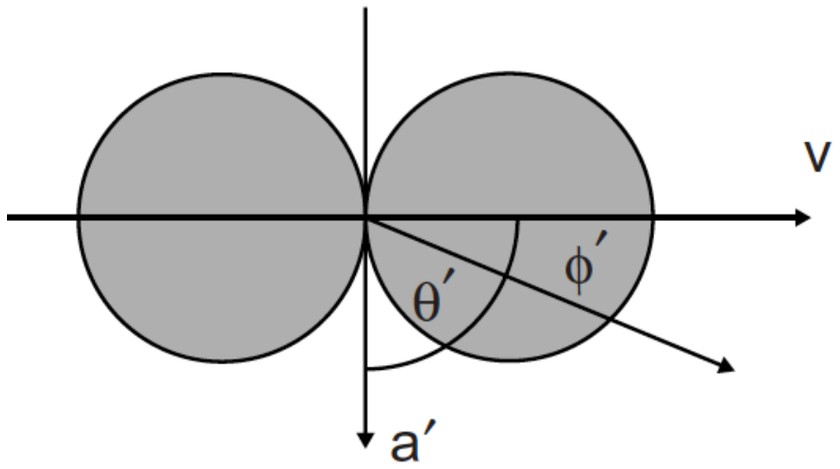
High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

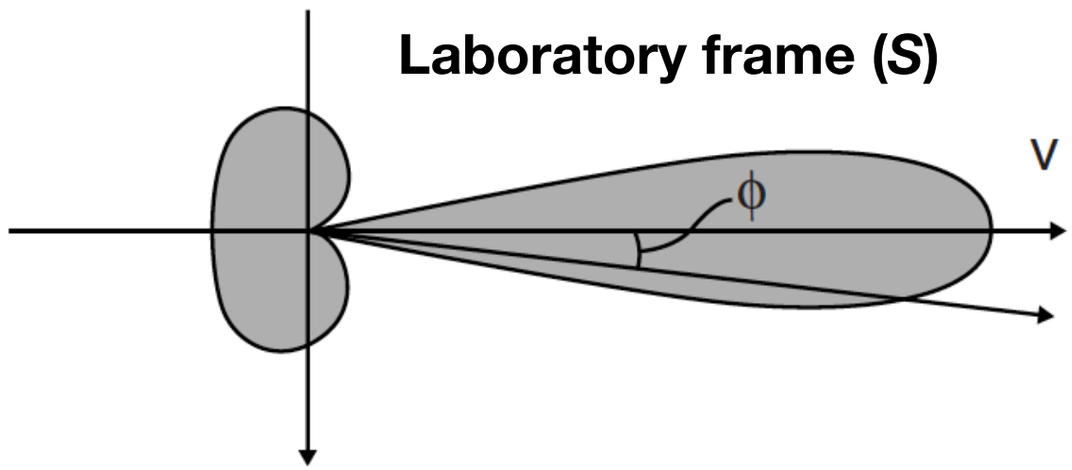
Emission pattern:

Electron's rest frame (*S'*)



To centre of particle's orbit

Laboratory frame (*S*)



To centre of particle's orbit

To illustrate the beaming of the radiation, consider the angles $\phi' = \pm\pi/4$, at which the intensity of radiation falls to half its maximum value, which occurs at $\phi' = \pi/2$ in the instantaneous rest frame of the electron. The corresponding angles in the laboratory frame of reference are

$$\sin\phi \approx \phi \approx \pm 1/\gamma \ ,$$

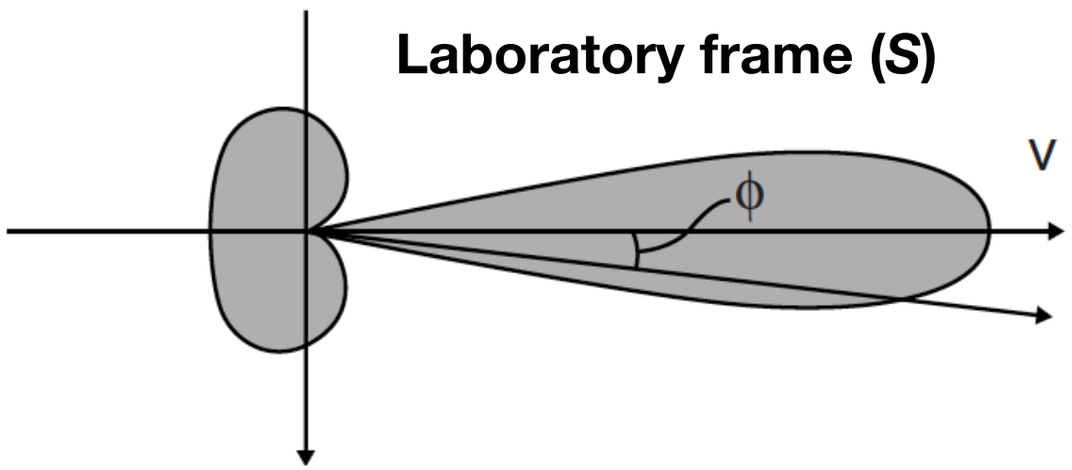
recalling that $\gamma \gg 1$. Thus, the radiation emitted within $-\pi/4 < \phi' < \pi/4$ is beamed in the direction of motion of the electron within the angles $-1/\gamma < \phi < 1/\gamma$. In the observer's frame S, the dipole beam pattern is very strongly elongated in the direction of motion of the electron

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Emission pattern:

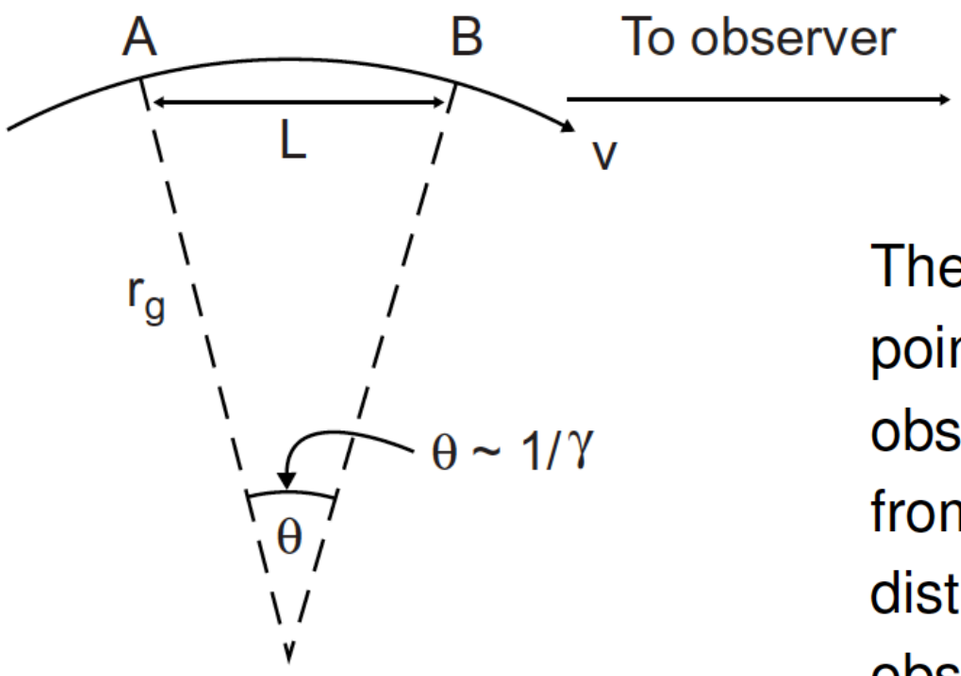


When this elongated beam pattern sweeps past the observer, a pulse of radiation is observed every time the electron's velocity vector lies within an angle of about $\pm 1/\gamma$ to the line of sight to the observer. The spectrum of the radiation received by the distant observer is the Fourier transform of this pulse from the time domain to the frequency domain.

Emission of synchrotron radiation

Synchrotron process

Pulse duration and frequency:



The observer sees significant radiation for only about $1/\gamma$ of the particle's orbit but the observed duration of the pulse is less than $1/\gamma$ times the period of the orbit because radiation emitted at the trailing edge of the pulse almost catches up with the radiation emitted at the leading edge.

The observer is located at a distance R from the point A. The radiation from A reaches the observer at time R/c. The radiation is emitted from B at time L/v later and then travels a distance $(R-L)$ at the speed of light to the observer. The trailing edge of the pulse therefore arrives at a time $L/v + (R-L)/c$. The duration of the pulse as measured by the observer is therefore

$$\Delta t = \left[\frac{L}{v} + \frac{(R-L)}{c}\right] - \frac{R}{c} = \frac{L}{v}\left(1 - \frac{v}{c}\right)$$

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Pulse duration and frequency:

$$\Delta t = \left[\frac{L}{v} + \frac{(R-L)}{c}\right] - \frac{R}{c} = \frac{L}{v}\left(1 - \frac{v}{c}\right)$$

The observed duration of the pulse is much less than the expected time interval *L/v*. Only if light propagated at an infinite velocity would the duration of the pulse be *L/v*.

We can write $$\frac{L}{v} = \frac{r_{\rm g}\theta}{v} \approx \frac{1}{\gamma\omega_{\rm r}} = \frac{1}{\omega_{\rm g}} ,$$

where $\omega_{\rm g}$ is the non-relativistic angular gyrofrequency and $\omega_r = \omega_{\rm g}/\gamma$ the relativistic angular gyrofrequency. We can also rewrite $(1 - v/c)$ as

$$\left(1 - \frac{v}{c}\right) = \frac{[1-(v/c)]\,[1+(v/c)]}{[1+(v/c)]} = \frac{\left(1 - v^2/c^2\right)}{1+(v/c)} \approx \frac{1}{2\gamma^2} ,$$

since $v \approx c$. Therefore, the observed duration of the pulse is $\boxed{\Delta t \approx \frac{1}{2\gamma^2\omega_{\rm g}} .}$

This means that the duration of the pulse as observed by a distant observer in the laboratory frame of reference is roughly $1/\gamma^2$ times shorter than the non-relativistic gyroperiod $T_{\rm g} = 2\pi/\omega_{\rm g}$.

Emission of synchrotron radiation

Synchrotron process

Pulse duration and frequency:

$$\text{Pulse duration, } \Delta t \approx \frac{1}{2\gamma^2 \omega_g} .$$

The maximum Fourier component of the spectral decomposition of the observed pulse of radiation is expected to correspond to a frequency $\nu \sim \Delta t^{-1}$, that is,

$$\boxed{\nu \sim \Delta t^{-1} \sim \gamma^2 \nu_g} ,$$

where ν_g is the non-relativistic gyrofrequency. This result is similar to the expression for the *critical frequency* (the frequency at which the emission peaks).

In the above calculation, it has been assumed that the electron moves in a circle about the magnetic field lines at pitch angle $\alpha = 90°$. The same calculation can be performed for any pitch angle with the result $\nu \sim \gamma^2 \nu_g \sin\alpha$.

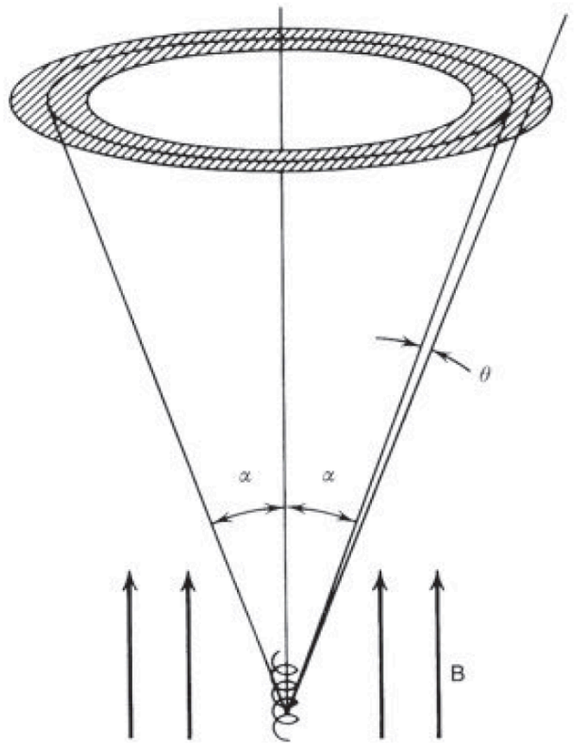
The reason for performing this simple exercise in detail is that the beaming of the radiation of ultra-relativistic electrons is a very general property and does not depend upon the nature of the force causing the acceleration.

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a single electron:

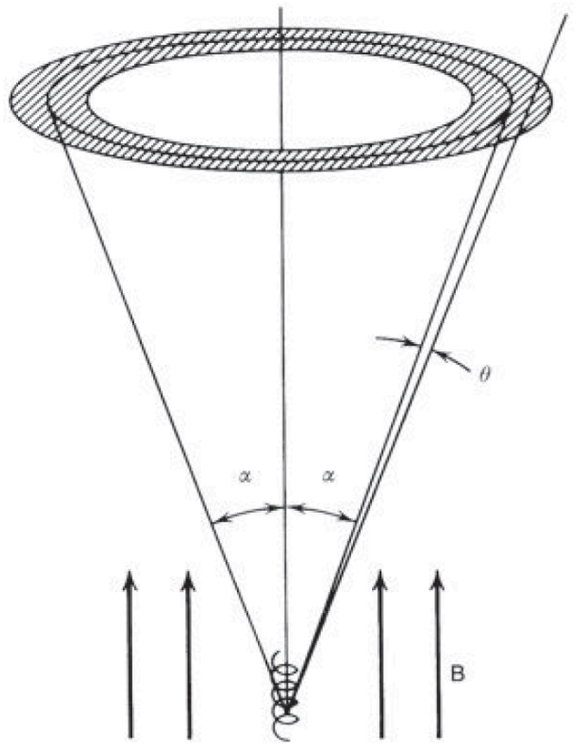
Synchrotron emission from an electron with pitch angle α. The radiation is confined to the shaded solid angle (Rybicki and Lightman, 1979).



High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process



Synchrotron spectrum from a single electron:

Synchrotron emission from an electron with pitch angle α. The radiation is confined to the shaded solid angle (Rybicki and Lightman, 1979).

There is no particularly simple way of deriving the spectral distribution of synchrotron radiation. The steps shared below follows from Rybicki and Lightman, 1979:

(i) Write down the expression for the energy emitted per unit bandwidth for an arbitrarily moving electron,
(ii) Select a suitable set of coordinates in which to work out the field components radiated by the electron spiralling in a magnetic field,
(iii) Then battle away at the algebra to obtain the spectral distribution of the field components.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

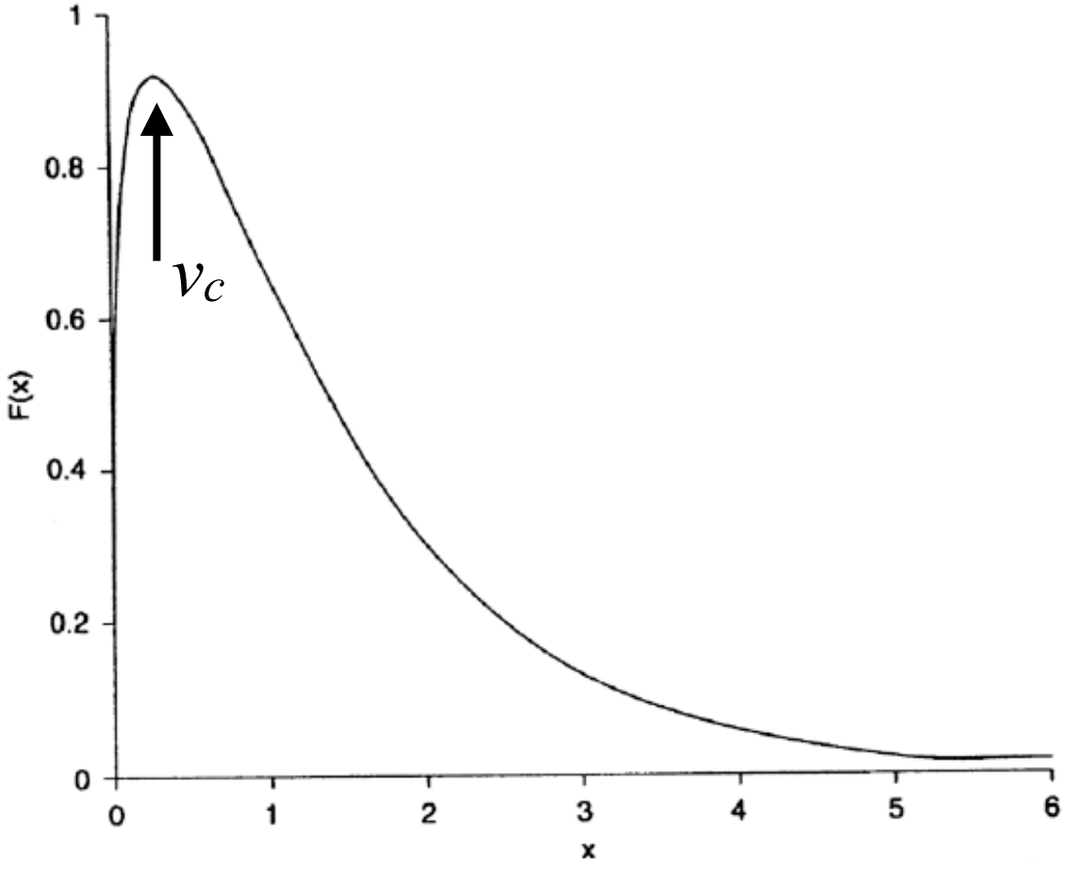
Synchrotron process

Synchrotron spectrum from a single electron:

The total emissivity of a single electron by synchrotron radiation, averaged over one orbit:

$$j(\omega) = \frac{\sqrt{3} e^3 B \sin\alpha}{8\pi^2 \epsilon_0 c m_e} F(x)$$

Here, emissivity is the total energy radiated per unit frequency interval per unit time.



$$x = \omega/\omega_C, \quad \omega_C = \frac{3}{2}\left(\frac{c}{v}\right)\gamma^3 \omega_r \sin\theta$$

$$F(x) = x\int_x^\infty K_{5/3}(z)\,\mathrm{d}z.$$

ω_C is known as the critical angular frequency. $K_{5/3}(z)$ is a modified Bessel function of order $5/3$. The form of this spectrum in terms of angular frequency ω is shown in the figure on the left in a in linear scale. It has a broad maximum centred roughly at the frequency $\nu \approx \nu_C$ with $\Delta\nu/\nu \sim 1$.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a single electron:

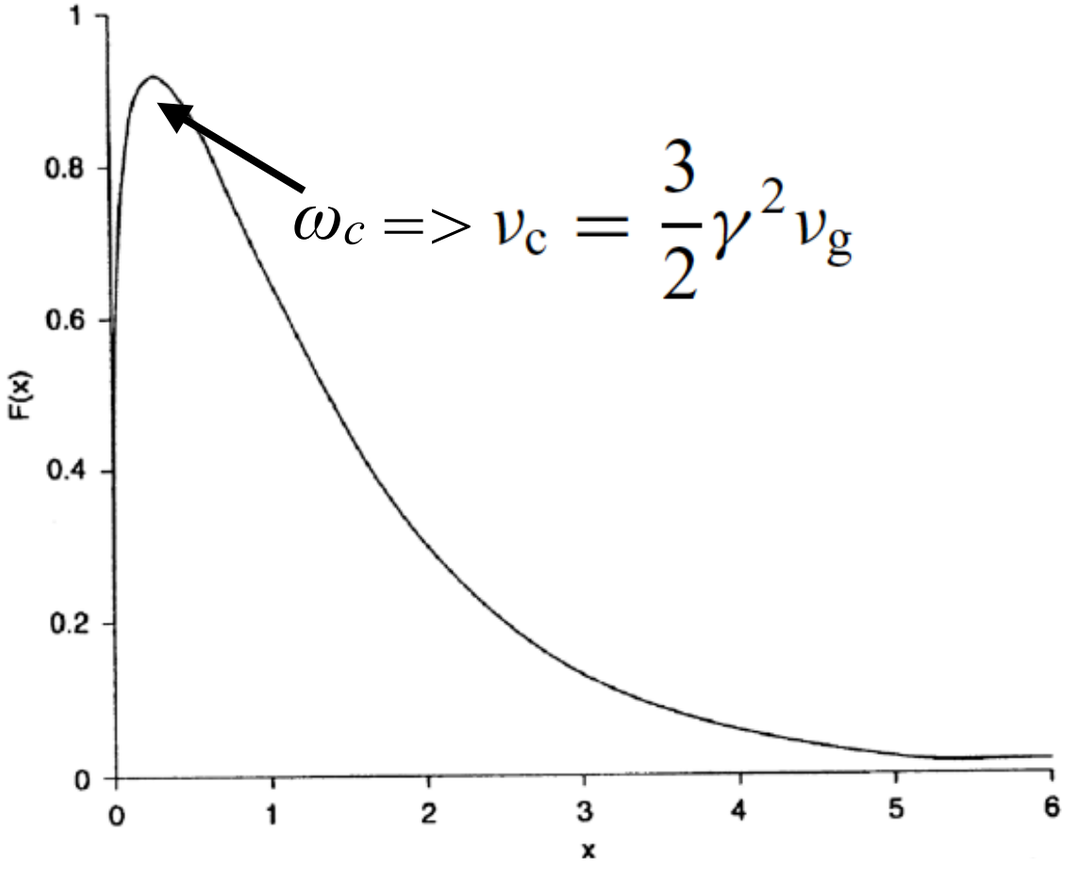
The total emissivity of a single electron by synchrotron radiation, averaged over one orbit:

$$j(\omega) = \frac{\sqrt{3} e^3 B \sin\alpha}{8\pi^2 \epsilon_0 c m_e} F(x)$$

Here, emissivity is the total energy radiated per unit frequency interval per unit time.

$$x = \omega/\omega_C, \quad \omega_C = \frac{3}{2}\left(\frac{c}{v}\right)\gamma^3 \omega_r \sin\theta$$

$$F(x) = x\int_x^\infty K_{5/3}(z)\,\mathrm{d}z.$$



ω_C is known as the critical angular frequency. $K_{5/3}(z)$ is a modified Bessel function of order $5/3$. The form of this spectrum in terms of angular frequency ω is shown in the figure on the left in a in linear scale. It has a broad maximum centred roughly at the frequency $\nu \approx \nu_C$ with $\Delta\nu/\nu \sim 1$.

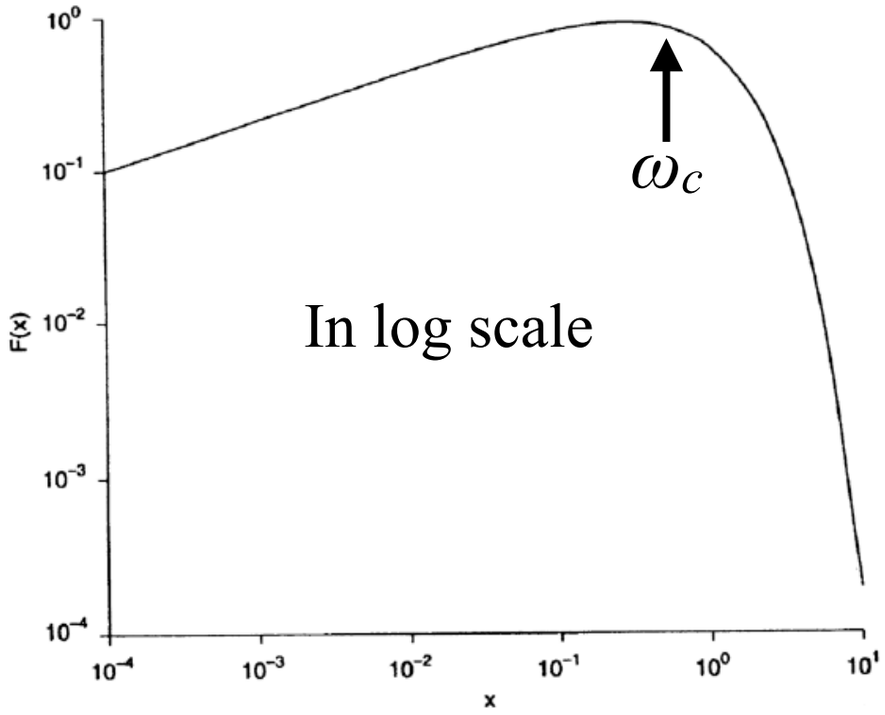
This critical frequency depends on the square of the electron energy and the bending radius. (1/2) of the power is radiated above this wavelength and (1/2) below.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a single electron:



At very high frequencies ($\nu >> \nu_c$), the emissivity spectrum from a single electron follows: $j(\nu) \propto \nu^{1/2} \exp(-\nu/\nu_c)$, which is dominated by the exponential cut-off.

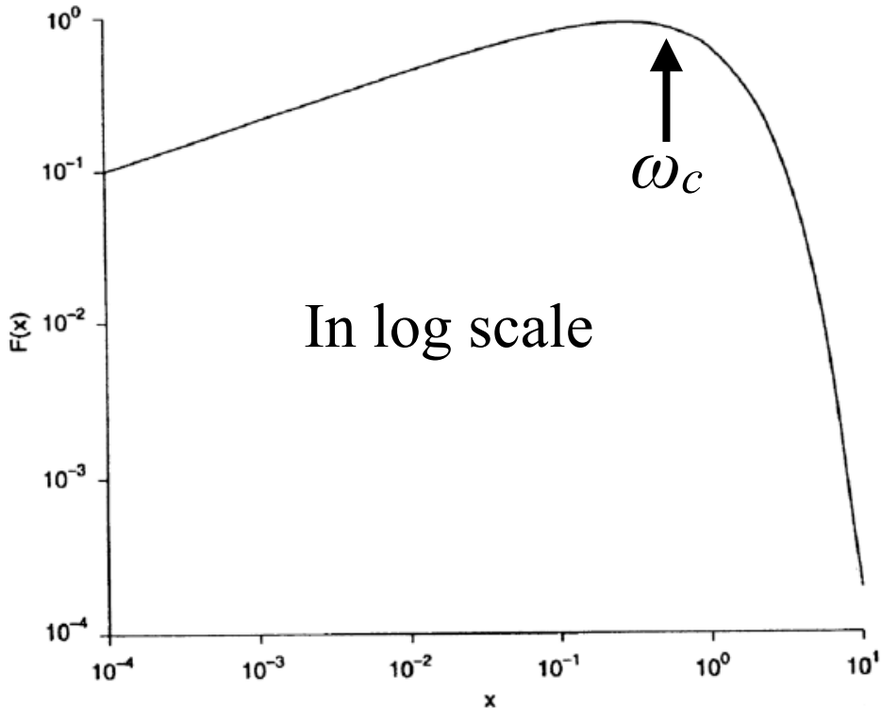
There is very little power at frequencies $\nu > \nu_c$ because there is very little structure in the polar diagram of the radiation emitted by the electron at angles $\theta << 1/\gamma$.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a single electron:



At very high frequencies ($\nu >> \nu_c$), the emissivity spectrum from a single electron follows: $j(\nu) \propto \nu^{1/2} \exp(-\nu/\nu_c)$, which is dominated by the exponential cut-off.

There is very little power at frequencies $\nu > \nu_c$ because there is very little structure in the polar diagram of the radiation emitted by the electron at angles $\theta << 1/\gamma$.

$$\text{At } \nu \ll \nu_c,\ j(\omega) = \frac{\sqrt{3} e^3 B \sin\alpha}{8\pi^2 \epsilon_0 c m_e} \frac{4\pi}{\sqrt{3}\Gamma(1/3)} \left(\frac{\omega}{2\omega_c}\right)^{1/3} = \frac{e^2}{3^{1/3}\Gamma(1/3) 2\pi\epsilon_0 c} \left(\frac{eB\sin\alpha}{\gamma m_e}\right)^{2/3} \omega^{1/3}$$

that is, the emissivity is proportional to $\nu^{1/3}$.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a power-law spectrum of electrons:

The emitted spectrum of electrons of energy E is quite sharply peaked near the critical frequency ν_{C} and is very much narrower than the breadth of the electron energy spectrum. Therefore, to a good approximation, it may be assumed that all the radiation of an electron of energy E is radiated at the critical frequency ν_{C} which we may approximate by

$$\nu \approx \nu_{\mathrm{C}} \approx \gamma^2 \nu_{\mathrm{g}} = \left(\frac{E}{m_{\mathrm{e}} c^2}\right)^2 \nu_{\mathrm{g}}; \qquad \nu_{\mathrm{g}} = \frac{eB}{2\pi m_{\mathrm{e}}}.$$

Therefore, the energy radiated in the frequency range ν to $\nu + \mathrm{d}\nu$ can be attributed to electrons with energies in the range E to $E + \mathrm{d}E$, which we assume to have power-law form $N(E) = \kappa E^{-p}$. We may therefore write

$$J(\nu)\,\mathrm{d}\nu = \left(-\frac{\mathrm{d}E}{\mathrm{d}t}\right) N(E)\,\mathrm{d}E.$$

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a power-law spectrum of electrons:

We have, $E = \gamma m_e c^2 = \left(\frac{\nu}{\nu_g}\right)^{1/2} m_e c^2, \qquad dE = \frac{m_e c^2}{2\nu_g^{1/2}} \nu^{-1/2} d\nu,$

and

$$-\left(\frac{dE}{dt}\right) = \tfrac{4}{3}\sigma_T c \left(\frac{E}{m_e c^2}\right)^2 \frac{B^2}{2\mu_0}.$$

Substituting these quantities, the emissivity may be expressed in terms of B, ν and fundamental constants.

$$J(\nu) = (\text{constants}) \quad B^{(p+1)/2} \nu^{-(p-1)/2}.$$

If the electron energy spectrum has power law index p, the spectral index of the synchrotron emission of these electrons, defined by $J(\nu) \propto \nu^{-\alpha}$, is $\alpha = (p-1)/2$. The spectral shape is determined by the shape of the electron energy spectrum rather than by the shape of the emission spectrum of a single particle. The quadratic nature of the relation between emitted frequency and the energy of the electron accounts for the difference in slopes of the emission spectrum and the electron energy spectrum.

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a power-law spectrum of electrons:

$$J(\nu) = (\text{constants}) \quad B^{(p+1)/2}\nu^{-(p-1)/2}.$$

We can write the energy spectrum in terms of intensity as a function of the photon energy E_γ as follows:

$$I(E_\gamma)dE_\gamma = \frac{J(\nu)d\nu}{\nu}$$

$$=> I(E_\gamma)dE_\gamma \propto \nu^{-\frac{p-1}{2}-1}d\nu \propto \nu^{-\frac{p+1}{2}}d\nu$$

$$\boxed{=> I(E_\gamma) \propto E_\gamma^{-\frac{(p+1)}{2}}} \quad \text{since } E_\gamma = h\nu$$

Similar to the inverse compton spectrum!

High Energy Astrophysics, Longair: Page 193-214

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a power-law spectrum of electrons: A simpler approach

The photons emitted as synchrotron radiation have a characteristic frequency given by

$$\omega_c = \frac{3}{2}\,\gamma^2 \left(\frac{eH}{mc}\right)$$

The synchrotron effect can be thought of as the interaction of the electron with "virtual" photons of the magnetic field. Denoting these photons by the subscript v, it is found (Jones, 1968) that they have an average energy

$$\epsilon_v = \frac{8}{5\pi\sqrt{3}} \left(\frac{\hbar}{mc}\right) eH \qquad \text{and a number density} \quad n_v = \frac{H^2}{8\pi\epsilon_v}$$

The resultant γ-rays have a characteristic energy found from equation

$$E_{\gamma,c} = \frac{3}{2}\,\gamma^2 \left(\frac{\hbar}{mc}\right) eH$$

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 155-157)

Emission of synchrotron radiation

Synchrotron process

Synchrotron spectrum from a power-law spectrum of electrons: A simpler approach

Now, plugin into the following equation to obtain the synchrotron spectrum,

$$q(E_\gamma) = 4\pi n_\nu \int_{E_\gamma}^{\infty} I_e(E) dE \times \frac{d\sigma(E_\gamma|\epsilon_\nu, E)}{dE_\gamma}$$

where, $\frac{d\sigma(E_\gamma, E)}{dE_\gamma} \approx \sigma_T \delta\left(E_\gamma - E_{\gamma,c}\right)$ with $E_{\gamma,c} = \frac{3}{2}\gamma^2 \left(\frac{\hbar}{mc}\right) eH$

For a power-law electron spectrum, $I_e \propto E^{-\Gamma}$ we get,

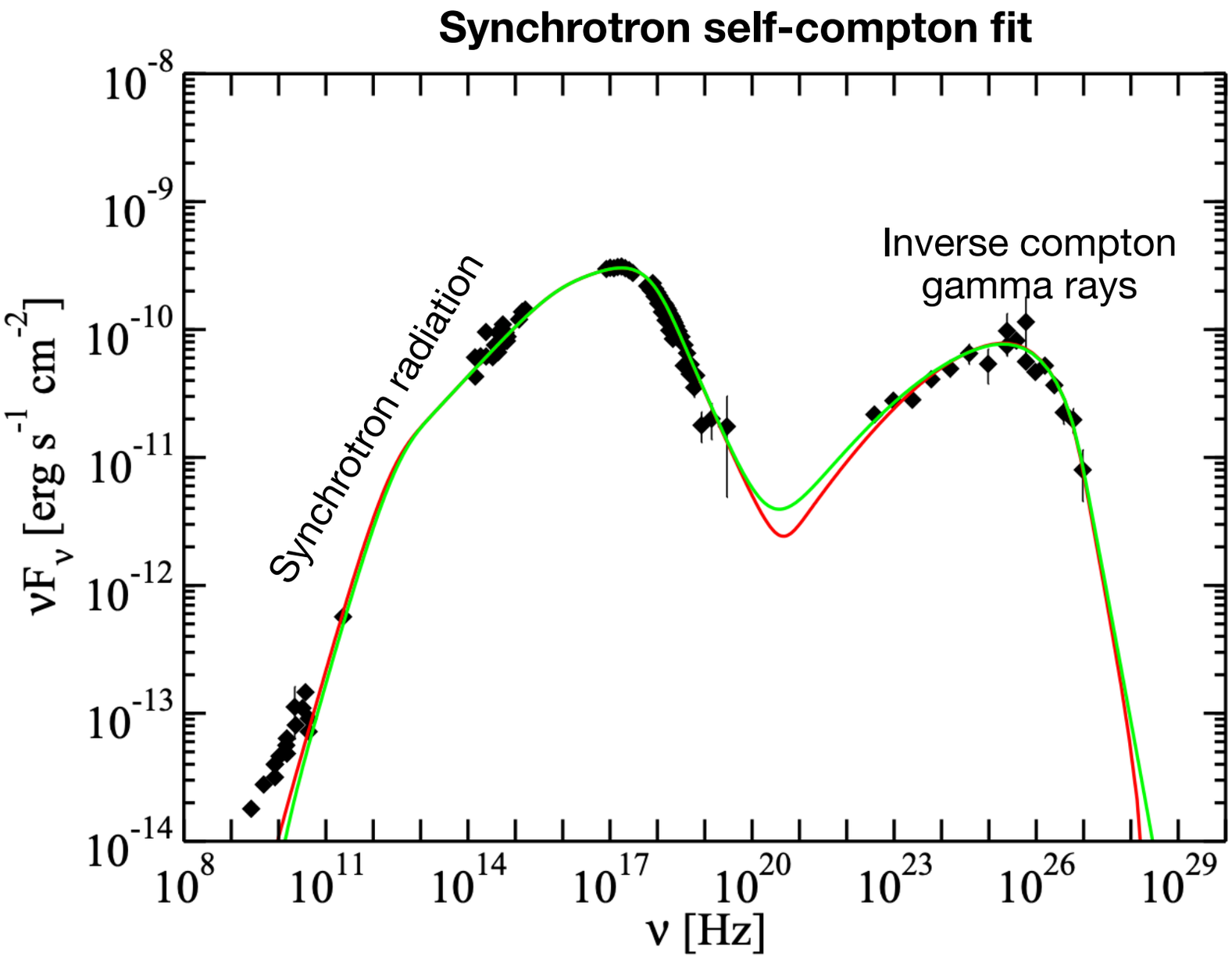
$$q(E_\gamma) \propto E_\gamma^{-\frac{(\Gamma+1)}{2}}$$

Cosmic gamma rays by Floyd William Stecker, NASA-SP-249 (1971), Document ID: 19710015288 (page 155-157)

Emission of synchrotron radiation

Synchrotron process

Multiwavelength spectrum from Mrk 421 (blazar):



Fermi Collaboration, https://arxiv.org/pdf/1106.1348.pdf

Next, we will discuss about cosmic-ray propagation.